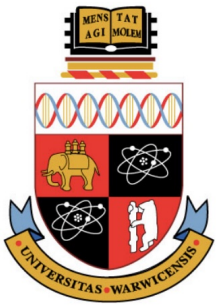
Market Making with Latent Volatility Regimes: An Avellaneda-Stoikov Framework with Hidden Markov Filtering

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Abstract

This dissertation extends the Avellaneda-Stoikov continuous-time market-making framework by introducing a latent Markov chain that jointly governs the mid-price volatility and the order-arrival intensity parameters. Whereas the existing partial-information model of Campi and Zabaljauregui (2020) allows the hidden regime to control only order-arrival intensities, the extension proposed here lets a single two-state chain simultaneously determine the diffusive volatility σ and the intensity parameters (A, k). The market maker maintains a real-time Bayesian belief over the current regime using a stochastic filter that updates on both price increments and order executions, and quotes a belief-weighted reservation price and optimal spread.

The model is calibrated empirically on Bitcoin-USDT perpetual-swap data from Binance. A two-state Gaussian hidden Markov model is fitted to 30-second realised volatility via the Baum-Welch algorithm, and intensity parameters are estimated separately for each regime from observed trade flow. Three strategies are compared through paired Monte Carlo simulation: a *naive* strategy that ignores regime structure and uses unconditional averages, a *partial-information* strategy that filters in real time, and an *oracle* strategy that observes the true regime.

On the primary calibration day (2024-03-30), the partial-information strategy outperforms the naive strategy in 83.5% of 200 paired simulation runs, delivering a mean PnL improvement of 6.4%. A robustness analysis across four consecutive trading days (2024-03-27 to 2024-03-30) confirms generalisability: the partial-information strategy achieves a mean win rate of 82.9% and a mean PnL improvement of 12.7%, with all improvements statistically significant at the 1% level. These results demonstrate that even approximate real-time regime detection, through stochastic filtering, materially improves market-making performance relative to ignoring latent structure.

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Notation and Abbreviations

Symbol / Abbreviation	Description
A-S	Avellaneda-Stoikov (framework)
BTC-USDT	Bitcoin-Tether perpetual swap pair
CARA	Constant Absolute Risk Aversion
HMM	Hidden Markov Model
KL	Kullback-Leibler (divergence)
LOB	Limit Order Book
MC	Monte Carlo
PnL	Profit and Loss
S_t	Mid-price at time t
Y_t	Latent regime (Markov chain state)
σ_i	Mid-price volatility in regime i
A_i, k_i	Intensity parameters in regime i
$\lambda(\delta)$	Fill intensity at quote distance δ
γ	Risk aversion parameter
q_t	Inventory at time t
π_t	Posterior belief vector over regimes
$\sigma^2_{\text{eff}}(\pi)$	Belief-weighted effective variance
$k_{\text{eff}}(\pi)$	Belief-weighted effective intensity decay
D_{KL}	Kullback-Leibler divergence
P	Transition probability matrix
T	Trading horizon
$\tau = T - t$	Time remaining
δ_t^+, δ_t^-	Ask and bid distances from mid-price
$\mathcal{F}_t$	Observable filtration at time t

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Chapter 1

Introduction

In modern financial markets, a market maker provides liquidity by posting continuous bid and ask quotes, earning the bid-ask spread as profit. With platforms such as Binance[1], anyone with exchange access can act as a market maker by submitting limit orders at chosen price levels.

The core challenge is managing two competing risks. *Inventory risk*: as the market maker fills incoming orders, they accumulate a position whose value fluctuates with the mid-price. *Execution risk*: wider quotes yield greater profit per trade but lower fill probability. Setting quotes optimally requires balancing these two forces in real time.

Ho and Stoll (1981) formulate dealer pricing as a stochastic dynamic programming problem in which the dealer maximises expected utility of terminal wealth under both return uncertainty and stochastic transaction arrivals, with optimal bid and ask quotes adjusting according to inventory. The foundational continuous-time treatment is due to Avellaneda and Stoikov (2008), who model the mid-price as an arithmetic Brownian motion, $dS_t = \sigma\, dW_t$, with incoming market orders arriving according to Poisson processes with intensities $\lambda(\delta) = A\exp(-k\delta)$. Under a CARA exponential utility objective, the optimal strategy is characterised by a reservation price $r = s - q\gamma\sigma^2(T-t)$ and an optimal spread $\delta^a + \delta^b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln(1+\gamma/k)$.

This framework, alongside subsequent extensions that improve tractability, allow more general intensity specifications, and incorporate richer inventory-risk objectives (Guéant et al., 2013; Guéant, 2017; Cartea and Jaimungal, 2015), has a key limitation: the assumption that market conditions are stationary. In reality, volatility regimes shift between calm and volatile periods, and order arrival rates change with them. Campi and Zabaljauregui (2020) address part of this limitation by introducing a hidden Markov chain Y_t that governs the order arrival intensities. The market maker cannot directly observe the

[1] `https://www.binance.com`

regime; they see only the mid-price evolution and their own order executions. This creates a stochastic filtering problem where the market maker maintains a belief π_t over the regime and uses it to set spreads. However, Campi and Zabaljauregui hold the mid-price volatility σ constant across regimes-the hidden chain affects only order arrival intensities, not the diffusive risk.

This dissertation extends the Avellaneda-Stoikov framework, building on Campi and Zabaljauregui (2020), by allowing the latent Markov chain to *jointly* govern both the mid-price volatility σ and the order arrival intensity parameters (A, k). Since the reservation price depends on σ^2 and the optimal spread depends on both σ^2 and k, a single hidden state controlling both creates a richer interaction between filtering and quoting. The market maker's belief π_t simultaneously adjusts the assessment of inventory risk (via $\mathbb{E}_\pi[\sigma^2]$) and execution probability (via the expected intensity parameters). This joint effect is absent in prior work: Campi and Zabaljauregui govern intensities alone while holding σ fixed, and Cartea and Jaimungal (2013) apply an HMM to price dynamics without letting it control intensities.

The empirical setting is the Bitcoin–USDT perpetual market on Binance. Cryptocurrency markets exhibit volatility clustering and evidence of latent regime changes in price dynamics (Ardia et al., 2019; Giudici and Abu Hashish, 2020). This makes them a natural testing ground for a regime-switching market-making model. Using one full day of limit order book and trade data (2024-03-30), a two-state hidden Markov model (Rabiner, 1989) is calibrated on 30-second realised volatility, yielding a calm regime ($\sigma_0 = 1.351$ USD$/\sqrt{\text{s}}$) and a volatile regime ($\sigma_1 = 2.819$ USD$/\sqrt{\text{s}}$), with a volatility ratio of approximately $2.1\times$.

Three strategies are compared via Monte Carlo simulation. The *naive* strategy ignores the regime and uses unconditional parameter averages. The *partial information* strategy runs an HMM filter in real time, updating its belief after each price increment and each order execution, and quotes using belief-weighted parameters. The *oracle* strategy observes the true regime directly and serves as an upper bound on achievable performance. The central finding is that the partial information strategy outperforms the naive strategy 83.5% of the time, with an average PnL improvement of 6.4%, and captures approximately 86% of the oracle's advantage. A robustness analysis across four consecutive trading days (2024-03-27 to 2024-03-30) confirms that these results are not specific to a single day: the partial information strategy achieves a mean win rate of 82.9% and a mean PnL improvement of 12.7% across all four days, with all improvements statistically significant at the 1% level. A further sensitivity analysis over the risk aversion parameter γ demonstrates that the advantage is robust to parameter choice and, in fact, increases with γ, consistent with the formal result (Proposition 3.5) that the naive strategy's spread underestimation grows with risk aversion.

1.1 Personal Motivation

My interest in market microstructure began during a summer internship at an investment bank, where I observed first-hand how electronic market makers continuously adjust their quotes in response to changing conditions. What struck me was the speed at which the market's character could shift: periods of orderly, low-volatility trading would give way to bursts of aggressive order flow in a matter of seconds. The idea that these shifts could be modelled as latent regime changes, and that a market maker could *learn* the current regime from observable data, became the seed of this project.

Several Warwick modules supported the development of this project. MA359 (Measure Theory) provided the formal measure-theoretic language underlying the filtering framework, including filtrations, conditional expectation, and the Radon–Nikodym theorem. MA3K1 (Mathematics of Machine Learning) contributed background in statistical learning and optimisation relevant to calibration. CS301 (Complexity of Algorithms) informed the computational perspective on tractability, while CS331 (Neural Computing) provided broader context for data-driven modelling.

The choice of the BTC-USDT perpetual on Binance was motivated by the accessibility of high-frequency cryptocurrency data and by the pronounced volatility clustering in these markets, which makes regime-switching models particularly relevant. The project connects several areas I find compelling: stochastic processes, Bayesian inference, and the practical challenge of making real-time decisions under uncertainty.

1.2 Contributions

The main contributions of this dissertation are as follows.

1. **Joint regime dependence of σ and (A, k).** The Avellaneda-Stoikov framework is extended so that a single latent Markov chain simultaneously governs both mid-price volatility and order-arrival intensity parameters, unlike Campi and Zabaljauregui (2020) where only intensities are regime-dependent.

2. **Two-channel Bayesian filter.** A discrete-time predict-update filter is developed that fuses both price-increment and order-execution observations, providing faster and more robust regime inference than either channel alone (Proposition 3.1).

3. **Formal characterisation of the naive bias.** Proposition 3.5 provides a closed-form expression for the spread underestimation incurred by ignoring regime struc-

ture, with the bias growing with the squared difference in intensity parameters between regimes.

4. **Quantitative simulation evidence.** A 200-run paired Monte Carlo study, calibrated to BTC-USDT perpetual futures and validated across four trading days, demonstrates that the partial information strategy closes 86% of the gap to the oracle on the primary day, with all improvements significant at the 1% level. A further robustness experiment under the raw empirical transition matrix confirms that the advantage is not an artefact of the rescaled switching design.

1.3 Outline

The remainder of this dissertation is organised as follows. Chapter 2 reviews the relevant literature, presenting the mathematical frameworks of Avellaneda and Stoikov (2008) and Campi and Zabaljauregui (2020) in detail. Chapter 3 develops the extended model, deriving the modified reservation price and spread formulae. Chapter 4 describes the empirical calibration procedure. Chapter 5 presents the Monte Carlo simulation results. Chapter 6 discusses the findings, their limitations, and directions for future work.

Chapter 2

Literature Review

In this chapter, we look more into the theoretical foundations on which the dissertation is built. Section 2.1 briefly reviews the early market microstructure literature on dealer pricing. Section 2.2 presents the Avellaneda-Stoikov (2008) optimal market making framework in detail, including the full derivation of the reservation price and optimal spread. Section 2.3 discusses the extensions that gave more detail to inventory management and spread calibration. Section 2.4 shows the Campi-Zabaljauregui (2020) partial information extension, which includes the hidden Markov model, the stochastic filter, the value function decomposition, and the resulting HJB equation. Section 2.5 reviews the application of hidden Markov models to cryptocurrency markets. Section 2.6 identifies the gap in the literature that motivates this dissertation.

2.1 Early Market Microstructure

Ho and Stoll (1981) establish the inventory-based view of dealer pricing: using stochastic dynamic programming, they show optimal quotes depend on inventory, return variance, and transaction arrivals, with dealers adjusting quotes to manage risk. Biais et al. (2005) survey broader microstructure including inventory costs, adverse selection, and price impact.

2.2 The Avellaneda-Stoikov Framework

Avellaneda and Stoikov (2008) develop a continuous-time limit order book model combining utility-theoretic dealer pricing with econophysics insights on order book microstructure. We present the full framework below.

2.2.1 Mid-Price Dynamics

The mid-price of the asset is assumed to follow an arithmetic Brownian motion,

$$dS_t = \sigma \, dW_t, \tag{2.1}$$

where W_t is a standard Wiener process and $\sigma > 0$ is a constant volatility parameter. There is no drift parameter, which reflects the assumption that the market maker has no directional view on the asset's price. This mid-price is used to value the market maker's inventory at terminal time; they cannot trade at S_t directly, but instead post limit orders at a distance from it.

2.2.2 The Frozen Inventory Value Function

Before introducing trading, consider a market maker holding a fixed inventory of q shares until terminal time T, maximising expected CARA utility of terminal wealth:

$$v(x, s, q, t) = E_t[-\exp(-\gamma(x + qS_T))] \,, \tag{2.2}$$

where x is cash, $s = S_t$ the current mid-price, and $\gamma > 0$ the risk aversion coefficient. Since $S_T \mid S_t = s$ is normally distributed with mean s and variance $\sigma^2(T - t)$, the moment-generating function gives

$$v(x, s, q, t) = -\exp(-\gamma x)\exp(-\gamma q s)\exp\left(\frac{\gamma^2 q^2 \sigma^2 (T - t)}{2}\right). \tag{2.3}$$

The value function factorises into cash, mark-to-market, and a penalty term that grows with q^2, capturing the increasing risk of holding large positions.

2.2.3 Reservation Prices

The reservation (or indifference) bid price r^b is defined as the price at which the market maker is indifferent between their current portfolio and a portfolio with one additional share purchased at price r^b:

$$v(x - r^b, s, q + 1, t) = v(x, s, q, t). \tag{2.4}$$

Symmetrically, the reservation ask price r^a satisfies

$$v(x + r^a, s, q - 1, t) = v(x, s, q, t). \tag{2.5}$$

Substituting the explicit form of v and simplifying yields

$$r^b(s,q,t) = s + (1-2q)\frac{\gamma\sigma^2(T-t)}{2} = s - (2q-1)\frac{\gamma\sigma^2(T-t)}{2}, \tag{2.6}$$

$$r^a(s,q,t) = s + (-1-2q)\frac{\gamma\sigma^2(T-t)}{2} = s - (2q+1)\frac{\gamma\sigma^2(T-t)}{2}. \tag{2.7}$$

The *reservation* or *indifference price* is defined as the average of these two,

$$r(s,q,t) = \frac{r^a + r^b}{2} = s - q\gamma\sigma^2(T-t). \tag{2.8}$$

When the market maker is long ($q > 0$), the reservation price lies below the mid-price, reflecting a desire to reduce inventory; when short ($q < 0$), it lies above. The adjustment is proportional to q, γ, and $\sigma^2(T-t)$.

2.2.4 Order Arrival Intensities

The market maker posts a bid at price $p^b = s - \delta^b$ and an ask at price $p^a = s + \delta^a$, where $\delta^b, \delta^a > 0$ are the distances from the mid-price. Market buy orders and market sell orders arrive according to independent Poisson processes N_t^a and N_t^b with intensities $\lambda^a(\delta^a)$ and $\lambda^b(\delta^b)$ respectively, where the intensities are decreasing functions of the distances.

Avellaneda and Stoikov derive a functional form by combining two empirical observations from the econophysics literature: market order sizes follow a power law $f^Q(x) \propto x^{-1-\alpha}$ with $\alpha \approx 1.5$ (Gabaix et al., 2003), and temporary price impact satisfies $\Delta p \propto \ln(Q)$ (Bouchaud et al., 2002). The probability that an order reaches distance δ is

$$\begin{aligned}
\lambda(\delta) &= \Lambda\,\mathbb{P}(\Delta p > \delta) \\
&= \Lambda\,\mathbb{P}(\ln(Q) > K\delta) \\
&= \Lambda \int_{\exp(K\delta)}^{\infty} x^{-1-\alpha}\,dx \\
&= A\exp(-k\delta),
\end{aligned} \tag{2.9}$$

where $A = \Lambda/\alpha$ and $k = \alpha K$. This exponential intensity model, $\lambda(\delta) = Ae^{-k\delta}$, is the standard specification used throughout the Avellaneda-Stoikov literature. The parameter A controls the baseline arrival rate (the intensity when the quote is exactly at the mid-price), and k controls the sensitivity of fill probability to the distance.

2.2.5 The HJB Equation and Ansatz

With trading, the market maker's value function is

$$u(s,x,q,t) = \max_{\delta^a,\delta^b} E_t[-\exp(-\gamma(X_T + q_T S_T))]\,. \tag{2.10}$$

By the dynamic programming principle, u satisfies the HJB equation:

$$\begin{cases} u_t + \frac{1}{2}\sigma^2 u_{ss} + \max_{\delta^b} \lambda^b(\delta^b)\big[u(s,x-s+\delta^b,q+1,t) - u\big] \\ \quad + \max_{\delta^a} \lambda^a(\delta^a)[u(s,x+s+\delta^a,q-1,t) - u] = 0, \\ u(s,x,q,T) = -\exp(-\gamma(x+qs)). \end{cases} \tag{2.11}$$

The *ansatz* $u(s,x,q,t) = -\exp(-\gamma x)\exp(-\gamma\theta(s,q,t))$ separates the value function, eliminating the dependence on x and yielding a PDE for θ:

$$\begin{cases} \theta_t + \frac{1}{2}\sigma^2\theta_{ss} - \frac{1}{2}\sigma^2\gamma\theta_s^2 \\ \quad + \max_{\delta^b}\left[\frac{\lambda^b(\delta^b)}{\gamma}\Big(1 - e^{\gamma(s-\delta^b-r^b)}\Big)\right] + \max_{\delta^a}\left[\frac{\lambda^a(\delta^a)}{\gamma}\big(1 - e^{-\gamma(s+\delta^a-r^a)}\big)\right] = 0, \\ \theta(s,q,T) = qs, \end{cases} \tag{2.12}$$

where $r^b(s,q,t) = \theta(s,q+1,t) - \theta(s,q,t)$ and $r^a(s,q,t) = \theta(s,q,t) - \theta(s,q-1,t)$. Note that the reservation prices from the frozen inventory problem reappear naturally in the PDE for θ. The first-order optimality conditions from the maximisations in Equation (2.12) give the implicit relations

$$s - r^b = \delta^b - \frac{1}{\gamma}\ln\left(1 - \gamma\frac{\lambda^b(\delta^b)}{\lambda^{b\prime}(\delta^b)}\right), \tag{2.13}$$

$$r^a - s = \delta^a - \frac{1}{\gamma}\ln\left(1 - \gamma\frac{\lambda^a(\delta^a)}{\lambda^{a\prime}(\delta^a)}\right). \tag{2.14}$$

2.2.6 Asymptotic Spread Formula

The PDE (2.12) does not admit a simple closed-form solution in general. However, for the symmetric exponential intensity $\lambda^a(\delta) = \lambda^b(\delta) = Ae^{-k\delta}$, an important simplification occurs. In this case, $\lambda/\lambda' = -1/k$, and the implicit optimality conditions (2.13)-(2.14) simplify to give the reservation prices exactly as in the frozen-inventory case. The optimal distances are then obtained by an asymptotic expansion of θ in the inventory variable q,

yielding the celebrated spread formula:

$$\delta^a + \delta^b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\Big(1+\frac{\gamma}{k}\Big). \tag{2.15}$$

The first term, $\gamma\sigma^2(T-t)$, is the *inventory risk premium*, increasing with risk aversion, volatility, and time. The second term, $\frac{2}{\gamma}\ln(1+\gamma/k)$, is the *execution risk premium*: when k is large, the market maker quotes tighter to attract fills; when k is small, wider spreads can be sustained. The individual optimal distances are

$$\delta^a = \frac{1}{2}\left(\gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\Big(1+\frac{\gamma}{k}\Big)\right) + (r^a - s), \tag{2.16}$$

$$\delta^b = \frac{1}{2}\left(\gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln\Big(1+\frac{\gamma}{k}\Big)\right) + (s - r^b), \tag{2.17}$$

so that the bid and ask are centred around the reservation price r rather than the mid-price s.

2.3 Extensions of the Avellaneda-Stoikov Framework

Guéant et al. (2013) solve with inventory constraints via ODE reduction; Guéant (2017) extend to general intensity functions; Cartea and Jaimungal (2015) use a quadratic running penalty $\zeta\int_0^T q_t^2\,dt$ as an alternative objective. Cartea and Jaimungal (2013) apply HMMs to tick-by-tick price dynamics with regime-dependent arrivals, but outside the A-S framework. Fodra and Labadie (2015) incorporate directional bets; Bayraktar and Ludkovski (2014) provide optimal liquidation frameworks extended by Campi and Zabaljauregui to partial information. Cartea et al. (2015) provides a broad treatment of algorithmic trading.

2.4 The Campi-Zabaljauregui Partial Information Framework

Campi and Zabaljauregui (2020) introduce partial information into the Avellaneda-Stoikov framework. Their key innovation is to let the order arrival intensities depend on a hidden Markov chain that the market maker cannot directly observe, thereby creating a combined problem of stochastic optimal control and stochastic filtering. We present their framework in detail below, specialised where appropriate to the two-state exponential intensity case that is most relevant for this dissertation.

2.4.1 Model Setup

The mid-price follows the same arithmetic Brownian motion as in Equation (2.1), with constant volatility σ. The market maker receives buy and sell market orders according to counting processes N^+ and N^- with stochastic intensities λ_t^+ and λ_t^-. These intensities depend on a hidden Markov chain Y_t with finite state space $\{1, \dots, k\}$, initial distribution μ_0, and continuous generator matrix $Q = (q_t^{ij})$. The intensity in regime i is $\Lambda_i^{\pm}(\delta) = a_i^{\pm} e^{-b_i^{\pm}\delta}$ for the exponential case.

The market maker observes W, N^+, and N^-, but *not* Y. They choose spread processes predictable with respect to the observable filtration $\mathbb{F}^{N^-,N^+,W}$, maximising

$$\sup_{\delta^-,\delta^+\in\mathcal{A}} \mathbb{E}\left[U_\gamma\left(X_T + S_T N_T - \ell(N_T) - \tfrac{1}{2}\sigma^2\zeta\int_0^T N_u^2\,du\right)\right], \tag{2.18}$$

where $U_\gamma(c) = (1 - e^{-\gamma c})/\gamma$, ℓ is a terminal execution cost, $\zeta \geq 0$ is a running inventory penalty, and $N_t = n_0 + N_t^- - N_t^+$ is the inventory process.

2.4.2 The Stochastic Filter

Since the market maker cannot observe Y_t, they form a *filter*: the conditional probability distribution of Y_t given their observable information. Define the optional projection

$$\Pi_t^i := \mathbb{P}(Y_t = i \mid \mathcal{F}_t^{N^-,N^+,W}), \quad 1 \leq i \leq k. \tag{2.19}$$

The vector $\Pi_t = (\Pi_t^1, \dots, \Pi_t^k)$ lives on the $(k-1)$-simplex $\Delta = \{\pi \in \mathbb{R}^k : \pi^i \geq 0, \sum_i \pi^i = 1\}$. Campi and Zabaljauregui show (Proposition 2.1.1) that the observable (filtered) intensities of $N^{\pm}$ are

$$\widehat{\lambda}_t^{\pm} = \sum_{i=1}^{k} \Pi_{t-}^i \Lambda_i^{\pm}(\delta_t^{\pm}), \tag{2.20}$$

where Π_{t-} denotes the left-limit (pre-jump) belief. The use of Π_{t-} rather than Π_t ensures that the filtered intensities are predictable, as required for the compensator of the counting processes.

The dynamics of the filter Π_t are given by the *Kushner-Stratonovich equations* (Proposi-

tion 2.1.2 of Campi and Zabaljauregui). For each component i,

$$\begin{aligned} d\Pi_t^i = &\sum_{j=1}^{k} \Big(q_t^{ji} \Pi_{t-}^j + \Pi_{t-}^i \Pi_{t-}^j \left(\mathbf{1}_t^-(\Lambda_j^- - \Lambda_i^-)(\delta_t^-) + \mathbf{1}_t^+(\Lambda_j^+ - \Lambda_i^+)(\delta_t^+) \right) \Big) \, dt \\ &+ \Pi_{t-}^i \left(\frac{\Lambda_i^-(\delta_t^-)}{\widehat{\Lambda}^-(\Pi_{t-}, \delta_t^-)} - 1 \right) dN_t^- + \Pi_{t-}^i \left(\frac{\Lambda_i^+(\delta_t^+)}{\widehat{\Lambda}^+(\Pi_{t-}, \delta_t^+)} - 1 \right) dN_t^+, \end{aligned} \tag{2.21}$$

where $\widehat{\Lambda}^{\pm}(\pi, \delta) = \sum_j \pi^j \Lambda_j^{\pm}(\delta)$ and all coefficients use the left-limit Π_{t-} to ensure predictability. Between arrivals, the belief drifts according to the chain's transition rates. At each fill, a Bayesian update shifts Π_t^i upward if the fill is more likely under regime i than the pre-jump average, and downward otherwise. In the two-state case, the filter reduces to a scalar $\pi_t = \Pi_t^1$, with dynamics driven by $N^{\pm}$.

2.4.3 Value Function Decomposition and Optimal Spreads

A central result of Campi and Zabaljauregui (2020) is the value function decomposition. They propose the ansatz

$$V(t, s, x, n, \pi) = U_\gamma(x + sn + \Theta(t, n, \pi)), \tag{2.22}$$

where Θ depends on time, inventory, and the filter state. This separates the diffusive component from the jump components, reducing the dimensionality. Theorem 3.1.1 of Campi and Zabaljauregui establishes this decomposition, while Theorem 3.1.4 shows that Θ is the unique continuous viscosity solution of a terminal-condition PIDE. In the two-state exponential intensity case, the resulting optimal spreads under partial information take the form

$$\widehat{\delta^{\pm}}(t, n, \pi) = \frac{1}{b} - \frac{2w(\pi)}{\hat{m}(\pi)} \Theta_\pi - \big(\Theta(t, n \mp 1, \pi/\hat{m}(\pi)) - \Theta(t, n, \pi) \big), \tag{2.23}$$

where $w(\pi) = (m-1)\pi(1-\pi)$ measures order-flow volatility and $\hat{m}(\pi)$ is the observable intensity ratio. The term $1/b$ is the risk-neutral spread; the remaining terms correct for regime uncertainty and inventory risk. Under full information, the market maker observes Y_t directly and the spreads reduce to closed-form expressions indexed by regime. Comparing the two reveals that the partially informed market maker quotes *wider* spreads to compensate for regime uncertainty.

2.5 Hidden Markov Models in Cryptocurrency Markets

Hidden Markov models have been widely applied to financial regime dynamics. Hamilton (1989) introduced the Markov-switching framework for nonstationary time series; Guidolin (2011) surveys subsequent applications. Rabiner (1989) provides the foundational tutorial on Baum-Welch estimation; see also Elliott et al. (2008) for a rigorous filtering-based treatment. In cryptocurrency markets, Ardia et al. (2019) find evidence of distinct volatility regimes in Bitcoin returns, Giudici and Abu Hashish (2020) model Bitcoin prices as switching between bull, stable, and bear regimes, and Pennoni et al. (2022) demonstrate shared regime structure across multiple cryptocurrencies. These findings motivate applying HMM-based regime switching to crypto market making, where pronounced volatility clustering makes constant-parameter assumptions particularly untenable.

2.6 The Literature Gap

The literature reviewed above reveals a clear gap within the Avellaneda-Stoikov market-making framework. The A-S framework and its extensions assume stationary market conditions: the volatility σ, the arrival rate parameter A, and the decay parameter k are all constant. Campi and Zabaljauregui (2020) relax the stationary assumption on the *intensities* by introducing a hidden Markov chain, but they hold σ *constant* across regimes. Cartea and Jaimungal (2013) apply a hidden Markov model to tick-by-tick price dynamics with regime-dependent trade arrivals and price revision distributions, but their model operates outside the A-S quote-based market-making framework and does not produce the reservation-price and optimal-spread structure that characterises the A-S literature.

Within the A-S framework, no existing work allows a single latent Markov chain to jointly govern both mid-price volatility and order arrival intensities. This coupling is economically natural, as volatile markets typically exhibit different order flow characteristics from calm ones, and mathematically consequential, since the reservation price depends on σ^2 while the optimal spread depends on both σ^2 and k. Introducing joint regime dependence creates a two-channel filtering effect that is richer than either channel alone. This is the gap that the present dissertation fills.

Chapter 3

Model

This chapter develops the extended market making model. The key innovation relative to Campi and Zabaljauregui (2020) is that a single latent Markov chain jointly governs both the mid-price volatility σ and the order arrival intensity parameters (A, k). Section 3.1 defines the market environment and the regime-switching dynamics. Section 3.2 presents the stochastic filter that the partially informed market maker uses to update her belief about the current regime. Section 3.3 derives the modified reservation price and optimal spread under the extended model. Section 3.4 defines the three quoting strategies: naive, partial information, and oracle, which will be compared in the simulation analysis.

Scope of the extension. The market environment (Brownian mid-price, CARA utility, exponential intensity) is inherited from Avellaneda and Stoikov (2008), and the HMM framework from Campi and Zabaljauregui (2020). The extensions are: (i) regime-dependent volatility (Section 3.1.1), (ii) two-channel predict-update filtering (Section 3.2), and (iii) belief-weighted quoting heuristic (Section 3.3), validated in Chapter 5.

Table 3.1 summarises the key notation used throughout the chapter.

3.1 Market Environment and Regime Dynamics

We retain the continuous-time limit order book setting of Avellaneda and Stoikov (2008) and Campi and Zabaljauregui (2020). The market maker trades a single asset over a finite horizon $[0, T]$ by posting bid and ask limit orders. Their state at time t is characterised by their cash position X_t, their inventory q_t, and the mid-price S_t.

Table 3.1: Summary of principal notation.

Symbol	Description
S_t	Mid-price at time t
q_t	Inventory (units held) at time t
X_t	Cash position at time t
Y_t	Latent regime (hidden Markov chain), $Y_t \in \{0, 1\}$
σ_i	Mid-price volatility in regime i
$A_i^{\pm}, k_i^{\pm}$	Intensity parameters (ask/bid) in regime i
P	Transition probability matrix of Y_t at step Δt
π_t	Posterior belief vector over regimes, $\pi_t^i = \mathbb{P}(Y_t = i \mid \mathcal{F}_t)$
σ_{eff}^2	Belief-weighted effective variance, $\sum_i \pi_t^i \sigma_i^2$
$k_{\text{eff}}^{\pm}$	Belief-weighted effective intensity decay
γ	Risk aversion parameter (CARA utility)
T	Trading horizon
$\delta_t^{\pm}$	Ask/bid quote distance from mid-price
$\lambda^{\pm}(\delta, i)$	Fill intensity at distance δ in regime i

3.1.1 Regime-Switching Mid-Price

Let Y_t be a continuous-time Markov chain on a finite state space $\mathcal{Y} = \{0, 1, \ldots, K-1\}$ with generator matrix $Q = (q^{ij})$. In the Avellaneda-Stoikov framework, the mid-price volatility σ is constant. In the Campi-Zabaljauregui framework, σ remains constant while only the intensity parameters are governed by the hidden chain. We extend this by allowing σ to depend on Y_t. The mid-price follows

$$dS_t = \sigma(Y_t)\, dW_t, \tag{3.1}$$

where W_t is a standard Wiener process independent of Y_t, and $\sigma : \mathcal{Y} \to \mathbb{R}_{>0}$ assigns a distinct volatility to each regime. The absence of drift reflects the same assumption as in Equation (2.1): the market maker has no directional view.

In the two-state case ($K = 2$), which is the focus of the simulation study, the chain alternates between a *calm* regime ($Y_t = 0$, volatility σ_0) and a *volatile* regime ($Y_t = 1$, volatility σ_1), with $\sigma_1 \gg \sigma_0$. The continuous-time dynamics are governed by the generator Q; at the simulation time step Δt, the transition probability matrix is $P(\Delta t) = e^{Q\Delta t}$, which we write simply as P:

$$P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}, \tag{3.2}$$

where $p_{ij} = \mathbb{P}(Y_{t+\Delta t} = j \mid Y_t = i)$. The stationary distribution $\boldsymbol{\mu} = (\mu_0, \mu_1)$ satisfies $\boldsymbol{\mu} P = \boldsymbol{\mu}$, giving $\mu_0 = p_{10}/(p_{01} + p_{10})$ and $\mu_1 = p_{01}/(p_{01} + p_{10})$.

3.1.2 Regime-Switching Order Arrival Intensities

The market maker posts a bid at $S_t - \delta_t^-$ and an ask at $S_t + \delta_t^+$. As in Section 2.2.4, market orders arrive via Poisson processes with exponential intensities, but the parameters now depend on Y_t:

$$\lambda^+(\delta^+, Y_t) = A^+(Y_t) \exp\big(-k^+(Y_t)\, \delta^+\big)\,, \tag{3.3}$$

$$\lambda^-(\delta^-, Y_t) = A^-(Y_t) \exp\big(-k^-(Y_t)\, \delta^-\big)\,, \tag{3.4}$$

where $A^+, A^-, k^+, k^- : \mathcal{Y} \to \mathbb{R}_{>0}$. We allow bid-ask asymmetry to capture the empirically observed directional imbalance in order flow.

3.1.3 Joint Regime Dependence

The critical modelling choice is that a *single* hidden chain Y_t simultaneously controls the volatility $\sigma(Y_t)$ and the intensity parameters $(A^\pm(Y_t), k^\pm(Y_t))$. This creates a joint dependence that is absent in prior work within the Avellaneda-Stoikov framework:

- Avellaneda and Stoikov (2008) and extensions: all parameters (σ, A, k) are constant.
- Campi and Zabaljauregui (2020): the intensity parameters (A, k) depend on Y_t, but σ is constant across regimes.
- Cartea and Jaimungal (2013): hidden states affect trade-arrival rates and price revision dynamics, but the model operates outside the A-S quote-based market-making framework.

The joint dependence is economically natural: when volatility increases, both the arrival rate and the sensitivity of fill probability to distance typically change. Mathematically, the hidden chain affects the problem through two channels:

1. The *inventory risk channel*: the reservation price depends on σ^2 (Equation 2.8), so the market maker's risk-adjusted valuation of the asset changes with the regime.
2. The *execution risk channel*: the optimal spread depends on k (Equation 2.15), so the trade-off between profit per trade and fill probability also changes with the regime.

A market maker who filters the regime therefore gains a two-channel advantage: simultaneously adjusting both inventory risk assessment and execution strategy.

3.1.4 Observable Information and Filtration

The chain Y_t is not directly observable. The market maker observes discrete-time mid-price increments sampled at grid points $t_n = n\Delta t$ and her fill processes N_t^+, N_t^-. Her information is the filtration

$$\mathcal{F}_t = \sigma\{S_{t_n} : t_n \leq t\} \vee \sigma\{N_u^+, N_u^- : 0 \leq u \leq t\}. \tag{3.5}$$

All trading decisions must be predictable with respect to $\mathcal{F}_t$. The discrete sampling is important: if the market maker observed the full continuous path of S, the quadratic variation $\langle S \rangle_t = \int_0^t \sigma^2(Y_u)\, du$ would reveal $\sigma(Y_t)$ and hence Y_t directly. By observing only discrete increments, the regime remains genuinely latent. Unlike Campi and Zabaljauregui (2020), where only fills carry regime information, here the mid-price increments are also informative through their variance.

3.2 The Stochastic Filter

The market maker maintains a belief vector $\pi_t = (\pi_t^0, \pi_t^1, \ldots, \pi_t^{K-1})$ over the state space $\mathcal{Y}$, where

$$\pi_t^i = \mathbb{P}(Y_t = i \mid \mathcal{F}_t), \quad i \in \mathcal{Y}. \tag{3.6}$$

The vector π_t lies on the $(K-1)$-simplex $\Delta = \{\pi \in \mathbb{R}^K : \pi^i \geq 0, \sum_i \pi^i = 1\}$.

3.2.1 Predict-Update Recursion

For the simulation, we use a discrete-time approximation of the continuous-time filter from Section 2.4.2. The time step Δt is chosen small enough that at most one fill occurs per step. The filter follows a predict-update recursion at each step.

Prediction step. Given the posterior π_{t-1} from the previous step, the one-step-ahead prediction under the Markov chain dynamics is

$$\bar{\pi}_t = \pi_{t-1}\, P, \tag{3.7}$$

where P is the transition matrix from Equation (3.2).

Update step. The predicted belief $\bar{\pi}_t$ is updated using the likelihood of the observed data at time t. The observation consists of two components: the price increment $\Delta S_t =$

$S_t - S_{t-1}$ and the trade event $e_t \in \{-1, 0, +1\}$, where $e_t = +1$ denotes an ask fill, $e_t = -1$ a bid fill, and $e_t = 0$ no fill.

For each regime i, the price likelihood is

$$L_i^{\text{price}} = \frac{1}{\sqrt{2\pi\sigma_i^2 \Delta t}} \exp\left(-\frac{(\Delta S_t)^2}{2\sigma_i^2 \Delta t}\right), \tag{3.8}$$

A large price increment is more likely under the volatile regime, shifting the belief accordingly. The trade event likelihood depends on the total intensity in regime i:

$$\Lambda_i = \lambda^+(\delta_t^+, i) + \lambda^-(\delta_t^-, i). \tag{3.9}$$

Then the event likelihood is

$$L_i^{\text{event}} = \begin{cases} e^{-\Lambda_i \Delta t}, & \text{if } e_t = 0, \\ \left(1 - e^{-\Lambda_i \Delta t}\right) \cdot \dfrac{\lambda^+(\delta_t^+, i)}{\Lambda_i}, & \text{if } e_t = +1, \\ \left(1 - e^{-\Lambda_i \Delta t}\right) \cdot \dfrac{\lambda^-(\delta_t^-, i)}{\Lambda_i}, & \text{if } e_t = -1. \end{cases} \tag{3.10}$$

Bayes' rule then computes the posterior:

$$\pi_t^i = \frac{\bar{\pi}_t^i \cdot L_i^{\text{price}} \cdot L_i^{\text{event}}}{\sum_{j \in \mathcal{Y}} \bar{\pi}_t^j \cdot L_j^{\text{price}} \cdot L_j^{\text{event}}}. \tag{3.11}$$

This two-channel update is the key difference in the discretised filter relative to Campi and Zabaljauregui (2020), where only L_i^{event} is informative (since constant σ makes the price likelihood identical across regimes).

3.2.2 Information Content of the Two Channels

To understand why the joint model produces a more informative filter, consider the two channels separately.

Price channel. The ratio of the price likelihoods between the volatile and calm regimes is

$$\frac{L_1^{\text{price}}}{L_0^{\text{price}}} = \frac{\sigma_0}{\sigma_1} \exp\left(\frac{(\Delta S_t)^2}{2\Delta t}\left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2}\right)\right). \tag{3.12}$$

When $\sigma_1 \gg \sigma_0$ (as in the empirical calibration), a large $|\Delta S_t|$ produces a very large likelihood ratio, strongly shifting the belief toward the volatile regime. Conversely, a small increment shifts the belief toward calm. This channel is absent in Campi and

Zabaljauregui's model.

Event channel. The event likelihood ratio depends on the regime-specific intensities. A fill at a wide spread is more likely in a high-intensity regime (which has larger A and/or smaller k), while no fill is more likely in a low-intensity regime. The event channel is the sole source of regime information in Campi and Zabaljauregui's framework.

Channel decomposition. The total per-step information gain of the joint filter can be decomposed additively in the log-likelihood domain. For the two-state case, the posterior log-odds update is

$$\log \frac{\pi_t^1}{\pi_t^0} = \log \frac{\bar{\pi}_t^1}{\bar{\pi}_t^0} + \underbrace{\log \frac{L_1^{\text{price}}}{L_0^{\text{price}}}}_{\text{price contribution}} + \underbrace{\log \frac{L_1^{\text{event}}}{L_0^{\text{event}}}}_{\text{event contribution}} . \tag{3.13}$$

If only the price contribution were used, the update would rely solely on return-volatility information; if only the event contribution, it would reduce to an execution-only channel as in Campi and Zabaljauregui's framework. The joint filter uses both additively, so even if one channel is weak on a given step, the other may still provide a strong signal. Since the two channels typically reinforce each other, the filter can converge faster to the true regime than either channel alone.

3.2.3 Filter Convergence Rate

The following proposition provides an idealised benchmark for how quickly the observation-update component of the filter can identify the true regime. It isolates the information gain from Bayesian updates alone, neglecting the transition drift introduced by the prediction step.

Proposition 3.1 (Observation-Update Convergence Rate)**.** *Let $Y = j$ be the true regime, held fixed over n consecutive observation times. Suppose the belief immediately before the first update satisfies $\bar{\pi}^j = \epsilon$ for some $\epsilon \in (0, 1)$, and consider n successive Bayesian updates (Equation 3.11) with conditionally independent observations,* without *intervening prediction steps. Then the expected posterior log-odds satisfy*

$$\mathbb{E}\left[\log \frac{\pi_n^j}{1 - \pi_n^j}\right] = \log \frac{\epsilon}{1 - \epsilon} + n \cdot D_{\mathrm{KL}}(f_j \,\|\, f_{1-j}), \tag{3.14}$$

where $D_{\mathrm{KL}}(f_j \| f_{1-j})$ is the KL divergence from the observation law in regime $1 - j$ to that in regime j, and f_i denotes the joint observation law (continuous price increment and

discrete trade event) in regime i.

Proof. The log-odds after one Bayesian update satisfy

$$\log \frac{\pi^j}{1-\pi^j} = \log \frac{\bar{\pi}^j}{1-\bar{\pi}^j} + \log \frac{L_j}{L_{1-j}},$$

where $L_i = L_i^{\text{price}} \cdot L_i^{\text{event}}$ is the combined likelihood. Taking expectations under the true regime j, the expected log-likelihood ratio is $\mathbb{E}_j[\log(L_j/L_{1-j})] = D_{\text{KL}}(f_j \| f_{1-j})$ by definition of the KL divergence. With no prediction step between updates, the posterior from one step becomes the prior for the next, and summing n conditionally independent updates gives the equality. □

Remark 3.2. *In the two-state case with Gaussian price increments, the KL divergence from the price channel alone is*

$$D_{\text{KL}}^{price}(f_j \| f_{1-j}) = \frac{1}{2}\left(\frac{\sigma_j^2}{\sigma_{1-j}^2} - 1 - \log \frac{\sigma_j^2}{\sigma_{1-j}^2}\right).$$

With the calibrated volatilities $\sigma_0 = 1.351$ and $\sigma_1 = 2.819$, the price-channel KL divergence when the true regime is volatile is $D_{\text{KL}}(f_1 \| f_0) \approx 0.94$ nats per step, and when the true regime is calm is $D_{\text{KL}}(f_0 \| f_1) \approx 0.35$ nats. Both are substantial, implying rapid convergence, though detecting the volatile regime is considerably faster than detecting the calm regime. By Proposition 3.1, the expected number of steps to reach 90% confidence from a 50% prior is approximately $\log(9)/0.94 \approx 2.3$ steps when the true regime is volatile and $\log(9)/0.35 \approx 6.3$ steps when calm. The empirical median convergence time of 15 steps (reported in Chapter 5) is longer because it includes the transition probability drag from the prediction step and because the fixed-regime assumption is only approximate.

3.3 Modified Reservation Price and Optimal Spread

The partially informed market maker cannot condition on the true regime Y_t. Instead, the market maker conditions on their belief π_t and computes *effective parameters* that aggregate across regimes.

3.3.1 Belief-Weighted Effective Volatility

Under full information, $r = s - q\gamma\sigma^2(T-t)$. Under partial information, the market maker replaces σ^2 with the belief-weighted expected variance:

$$\sigma^2_{\text{eff}}(\pi_t) = \mathbb{E}_{\pi_t}[\sigma^2(Y_t)] = \sum_{i\in\mathcal{Y}} \pi^i_t \, \sigma^2_i. \tag{3.15}$$

In the two-state case, $\sigma^2_{\text{eff}} = \pi^0_t\sigma^2_0 + \pi^1_t\sigma^2_1$. The modified reservation price under partial information is then

$$\hat{r}(s, q, t, \pi_t) = s - q\gamma \, \sigma^2_{\text{eff}}(\pi_t) \, (T-t). \tag{3.16}$$

When the belief is concentrated on a single regime, this reduces to the full-information reservation price. Equation (3.16) treats the belief as frozen over $[t, T]$; the exact future inventory risk involves $\mathbb{E}[\int_t^T \sigma^2(Y_u)\, du \mid \mathcal{F}_t]$, which depends on the chain's transition dynamics. The frozen-belief form $\sigma^2_{\text{eff}}(\pi_t)(T-t)$ is an approximation that becomes exact when the regime is absorbing or the horizon is short. The following proposition provides a least-squares justification for this aggregation.

Proposition 3.3 (Least-Squares Justification for Belief-Weighted Variance)**.** *Let $\pi \in \Delta$ be a belief over the regime space $\mathcal{Y}$, and let σ^2_i denote the mid-price variance in regime i. The local inventory risk penalty for a market maker holding q units over the remaining interval τ is*

$$R(\hat{\sigma}^2, q, \tau, \pi) = \mathbb{E}_\pi\left[\frac{\gamma}{2} q^2 (\sigma^2(Y) - \hat{\sigma}^2)^2 \tau^2\right] + \frac{\gamma}{2} q^2 \hat{\sigma}^4 \tau^2,$$

where $\hat{\sigma}^2$ is the volatility estimate used in the reservation price. The unique minimiser of the expected quadratic penalty $\mathbb{E}_\pi[(\sigma^2(Y) - \hat{\sigma}^2)^2]$ is

$$\hat{\sigma}^{2*} = \sigma^2_{\text{eff}}(\pi) = \sum_{i\in\mathcal{Y}} \pi^i \sigma^2_i.$$

Proof. The expected squared deviation is $g(\hat{\sigma}^2) = \sum_i \pi^i(\sigma^2_i - \hat{\sigma}^2)^2$. Taking the derivative with respect to $\hat{\sigma}^2$:

$$\frac{dg}{d\hat{\sigma}^2} = -2\sum_i \pi^i(\sigma^2_i - \hat{\sigma}^2) = -2\left(\sum_i \pi^i \sigma^2_i - \hat{\sigma}^2\right).$$

Setting this to zero gives $\hat{\sigma}^{2*} = \sum_i \pi^i \sigma^2_i = \sigma^2_{\text{eff}}(\pi)$. The second derivative $d^2g/d(\hat{\sigma}^2)^2 = 2 > 0$ confirms this is a minimum. □

Note that this is a frozen-belief approximation rather than a formal solution to the partially observed HJB equation (which would require solving a PIDE on the belief simplex, as in Campi and Zabaljauregui (2020)). However, Proposition 3.3 provides a formal

least-squares justification for this aggregation, and the Monte Carlo results in Chapter 5 demonstrate that it captures most of the value of regime awareness.

Remark 3.4 (Connection to Measure-Theoretic Foundations)**.** *The belief-weighted volatility $\sigma^2_{\text{eff}}(\pi_t) = \mathbb{E}_{\pi_t}[\sigma^2(Y_t)]$ is a conditional expectation with respect to the posterior measure π_t on the regime space $\mathcal{Y}$, viewed as a measurable space equipped with the counting measure. More precisely, the filter update in Equation* (3.11) *can be interpreted as a Radon-Nikodym derivative computation: the posterior π_t is absolutely continuous with respect to the prior $\bar{\pi}_t$ (provided $\bar{\pi}_t$ has strictly positive components), with density*

$$\frac{d\pi_t}{d\bar{\pi}_t}(i) = \frac{L_i^{price} \cdot L_i^{event}}{\sum_j \bar{\pi}_t^j \, L_j^{price} \cdot L_j^{event}},$$

which is precisely the normalised likelihood ratio. This discrete-time update is a discrete-time analogue of the normalised-likelihood viewpoint underlying continuous-time filtering and the Zakai equation (Elliott et al., 2008). The filtration $\mathcal{F}_t$ (Equation 3.5) is generated by the observable processes, and all trading decisions must be predictable with respect to it, ensuring that the market maker cannot use future information or unobservable regime realisations.

3.3.2 Belief-Weighted Effective Intensity Decay

The optimal spread depends on k through $\frac{2}{\gamma}\ln(1+\gamma/k)$. Under the spread approximation used here (where the per-side base spread is approximated as $1/k + \gamma\sigma^2\tau/2$), the natural aggregation is the expected inverse:

$$\frac{1}{k^{\pm}_{\text{eff}}(\pi_t)} = \mathbb{E}_{\pi_t}\left[\frac{1}{k^{\pm}(Y_t)}\right] = \sum_{i\in\mathcal{Y}} \pi_t^i \cdot \frac{1}{k_i^{\pm}}. \tag{3.17}$$

Since $1/k$ enters linearly in the approximate spread formula, the belief-weighted value $\mathbb{E}_\pi[1/k]$ is the appropriate aggregation rather than $1/\mathbb{E}_\pi[k]$.

The following proposition quantifies the bias introduced by the naive aggregation.

Proposition 3.5 (Spread Underestimation by the Naive Strategy)**.** *Let $k : \mathcal{Y} \to \mathbb{R}_{>0}$ assign a decay parameter to each regime, with $k_0 \neq k_1$ in the two-state case. For any non-degenerate belief π (i.e., $\pi^0, \pi^1 > 0$), the following strict inequality holds:*

$$\mathbb{E}_\pi\left[\frac{1}{k}\right] > \frac{1}{\mathbb{E}_\pi[k]}. \tag{3.18}$$

In the two-state case, the gap is exactly

$$\mathbb{E}_\pi\left[\frac{1}{k}\right] - \frac{1}{\mathbb{E}_\pi[k]} = \frac{\pi^0\pi^1(k_1-k_0)^2}{k_0k_1(\pi^0k_0+\pi^1k_1)}. \tag{3.19}$$

Consequently, the naive strategy underestimates the per-side base spread by this amount, and the bias grows with the squared difference $(k_1-k_0)^2$ *between regimes.*

Proof. The inequality (3.18) follows from Jensen's inequality applied to the strictly convex function $f(k) = 1/k$ on $\mathbb{R}_{>0}$, with strictness holding because π is non-degenerate and $k_0 \neq k_1$.

For the explicit gap, write $\bar{k} = \pi^0k_0 + \pi^1k_1$. Then

$$\mathbb{E}_\pi\left[\frac{1}{k}\right] - \frac{1}{\bar{k}} = \frac{\pi^0}{k_0} + \frac{\pi^1}{k_1} - \frac{1}{\bar{k}} = \frac{\pi^0k_1\bar{k} + \pi^1k_0\bar{k} - k_0k_1}{k_0k_1\bar{k}}.$$

Expanding the numerator using $\bar{k} = \pi^0k_0 + \pi^1k_1$ and simplifying:

$$\pi^0k_1(\pi^0k_0+\pi^1k_1) + \pi^1k_0(\pi^0k_0+\pi^1k_1) - k_0k_1 = \pi^0\pi^1(k_1^2+k_0^2) + (\pi^{02}+\pi^{12}-1)k_0k_1.$$

Since $\pi^0+\pi^1 = 1$, we have $\pi^{02}+\pi^{12} = 1-2\pi^0\pi^1$, and the expression reduces to $\pi^0\pi^1(k_1-k_0)^2$, giving the result. □

Operationally, the partially informed market maker who uses $\mathbb{E}[1/k]$ quotes wider spreads than a naive trader using $1/\mathbb{E}[k]$, compensating for regime uncertainty. Within the spread approximation used here, the naive strategy underestimates the per-side base spread by exactly Equation (3.19).

Remark 3.6 (Numerical illustration with calibrated parameters)**.** *Using the ask-side intensity parameters from the BTC-USDT calibration (Chapter 4),* $k_0^+ = 1.714$ *and* $k_1^+ = 0.937$*, with the stationary belief* $\pi = (2/3, 1/3)$*, the bias is*

$$\frac{(2/3)(1/3)(1.714-0.937)^2}{1.714\times 0.937\times((2/3)(1.714)+(1/3)(0.937))} \approx 0.057 \text{ USD.}$$

The naive per-side base spread from $1/\mathbb{E}[k^+]$ *is* $1/1.455 \approx 0.687$ *USD, while the belief-weighted value* $\mathbb{E}[1/k^+] \approx 0.744$ *USD. The underestimation of roughly 8% per side compounds across both bid and ask quotes and across many trades, contributing to the aggregate PnL gap observed in Chapter 5. Combined with the volatility-channel bias, the naive strategy uses* $\bar{\sigma}^2 = \sum_i \mu_i\sigma_i^2 \approx 3.87$ *instead of adjusting between* $\sigma_0^2 = 1.83$ *and* $\sigma_1^2 = 7.95$*—the total quoting error is substantial.*

3.3.3 Belief-Weighted Quoting Rule Under Partial Information

Combining the effective parameters, the per-side base spread under partial information is

$$\text{base}^{\pm}(\pi_t) = \frac{1}{k^{\pm}_{\text{eff}}(\pi_t)} + \frac{1}{2}\gamma\,\sigma^2_{\text{eff}}(\pi_t)\,\tau, \tag{3.20}$$

where $\tau = T - t$ is the remaining time. The inventory skew adjusts the bid and ask distances symmetrically around the reservation price:

$$\text{skew}(q, \pi_t) = \gamma\,\sigma^2_{\text{eff}}(\pi_t)\,q\,\tau. \tag{3.21}$$

This yields the following belief-weighted quoting rule:

$$\hat{\delta}^+_t = \text{base}^+(\pi_t) - \text{skew}(q_t, \pi_t), \tag{3.22}$$

$$\hat{\delta}^-_t = \text{base}^-(\pi_t) + \text{skew}(q_t, \pi_t), \tag{3.23}$$

subject to $\hat{\delta}^{\pm}_t \in [\delta_{\min}, \delta_{\max}]$. When $q_t > 0$, the ask tightens and the bid widens to shed inventory-the standard A-S mechanism, now with belief-dependent magnitudes. Hard inventory limits are also imposed:

$$\hat{\delta}^-_t = \delta_{\max} \text{ if } q_t \geq q_{\max}, \qquad \hat{\delta}^+_t = \delta_{\max} \text{ if } q_t \leq -q_{\max}. \tag{3.24}$$

3.4 Strategy Definitions

To isolate the value of regime awareness, we define three market making strategies that share the same market environment (mid-price, regime process, order flow) but differ in how they use information about the regime.

3.4.1 Naive Strategy

The naive market maker computes fixed parameters from the prior π_0 and uses them throughout without updating:

$$\bar{\sigma}^2 = \sum_{i \in \mathcal{Y}} \pi^i_0\,\sigma^2_i, \tag{3.25}$$

$$\bar{k}^{\pm} = \sum_{i \in \mathcal{Y}} \pi^i_0\,k^{\pm}_i. \tag{3.26}$$

Crucially, the naive strategy uses $1/\bar{k}^{\pm}$ rather than $\mathbb{E}_{\pi_0}[1/k^{\pm}]$, systematically underestimating the belief-weighted per-side base spread by Jensen's inequality (Equation 3.18).

Remark 3.7 (Relation to the Standard A-S Framework)**.** *The naive strategy reduces the regime-switching model to a constant-parameter Avellaneda-Stoikov-type benchmark: it uses fixed $(\bar{\sigma}, \bar{k})$ throughout, chosen as the unconditional averages under the prior. It inherits the belief-weighted quoting structure (including bid-ask asymmetry and hard inventory limits) rather than the exact closed-form A-S solution, but the key feature is the absence of regime adaptation. Thus, the comparison between the naive and partial information strategies in Chapter 5 can be interpreted as a test of the marginal value of regime-switching relative to a constant-parameter benchmark. Any improvement of the partial information strategy over the naive quantifies the cost of the stationarity assumption.*

3.4.2 Partial Information Strategy

At each step, the market maker: (1) observes ΔS_t and e_t; (2) updates π_t via Equations 3.7-3.11; (3) computes $\sigma^2_{\text{eff}}(\pi_t)$ and $1/k^{\pm}_{\text{eff}}(\pi_t)$; (4) sets quotes via Equations (3.22)-(3.23). This is the central contribution: by filtering both price and event observations, the market maker adapts to the inferred regime through both channels simultaneously.

3.4.3 Oracle Strategy

The oracle observes Y_t directly and uses exact regime-specific parameters:

$$\delta_t^{+,\text{oracle}} = \frac{1}{k^+(Y_t)} + \frac{1}{2}\gamma\sigma^2(Y_t)\,\tau - \gamma\sigma^2(Y_t)\,q_t\,\tau, \tag{3.27}$$

$$\delta_t^{-,\text{oracle}} = \frac{1}{k^-(Y_t)} + \frac{1}{2}\gamma\sigma^2(Y_t)\,\tau + \gamma\sigma^2(Y_t)\,q_t\,\tau. \tag{3.28}$$

This provides an upper bound: no strategy with the same model can do better.

3.4.4 Paired Simulation Design

To enable fair comparison, all three strategies are evaluated on the same set of simulated market environments. For each Monte Carlo trial $m \in \{1, \ldots, M\}$: (1) a single regime path $\{Y_t^{(m)}\}_{t=0}^{T/\Delta t}$ is drawn from the Markov chain; (2) each strategy is simulated on this common regime path with independent random number streams for the Brownian increments and order arrivals.

The common regime path ensures that performance differences reflect the quoting strategy, not luck in the regime draw. Independent Brownian and Poisson streams within each trial ensure that fill sequences differ across strategies, as they would in practice. This paired design reduces variance in the inter-strategy comparisons.

Chapter 4

Data and Calibration

This chapter describes the empirical data and the calibration procedure used to estimate the model parameters from Chapter 3. Section 4.1 describes the dataset. Section 4.2 presents the hidden Markov model calibration on realised volatility. Section 4.3 estimates the order arrival intensity parameters. Section 4.4 summarises the simulation configuration and discusses the choices made in translating the empirical estimates into parameters for the Monte Carlo experiments.

4.1 Dataset

The primary calibration uses one full trading day of limit order book data for the BTC-USDT perpetual futures contract on Binance[1], recorded on 30 March 2024 (Binance, 2024); additional days (2024-03-27 to 2024-03-29) are used for the robustness analysis in Section 5.7. The raw data consists of two streams: (1) order book snapshots, from which the mid-price S_t is computed as the average of the best bid and ask, and (2) trade-level data recording each executed trade with timestamp, price, size, and aggressor side. Figure 4.1 shows the mid-price (fluctuating around \$71,000) and the bid-ask spread, which appears to widen during more volatile episodes and narrow during calmer ones, suggesting time variation consistent with the regime-switching hypothesis.

[1] `https://www.binance.com`

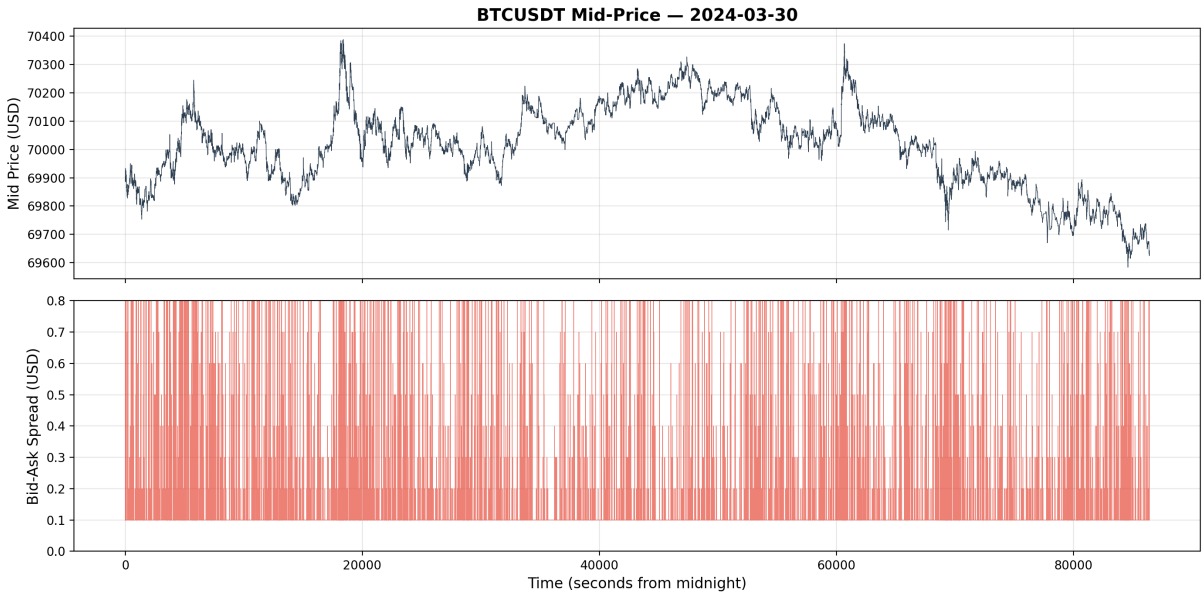


Figure 4.1: Mid-price and bid-ask spread for BTC-USDT on 2024-03-30.

4.2 HMM Calibration

4.2.1 Realised Volatility Construction

We compute realised volatility using 30-second rolling windows of 1-second mid-price returns, updated every 10 seconds (i.e. with a 10-second stride). Each window contains 30 return observations, and the realised volatility is the sample standard deviation within that window. The 30-second window balances reliability against the ability to detect regime transitions on the scale of tens of seconds; the 10-second stride determines the HMM time step.

Approximately 3.9% of windows have realised volatility below \$0.01 (near-unchanged mid-price). Because the log transform is unstable near zero, these windows are excluded before fitting the HMM and assigned to the calm regime post hoc as a pragmatic preprocessing choice.

Figure 4.2 plots the 30-second realised volatility series alongside the HMM-inferred regime classification. The time series displays clear clustering: extended periods of low volatility alternate with bursts of much higher volatility, supporting the use of a two-state regime model for this dataset.

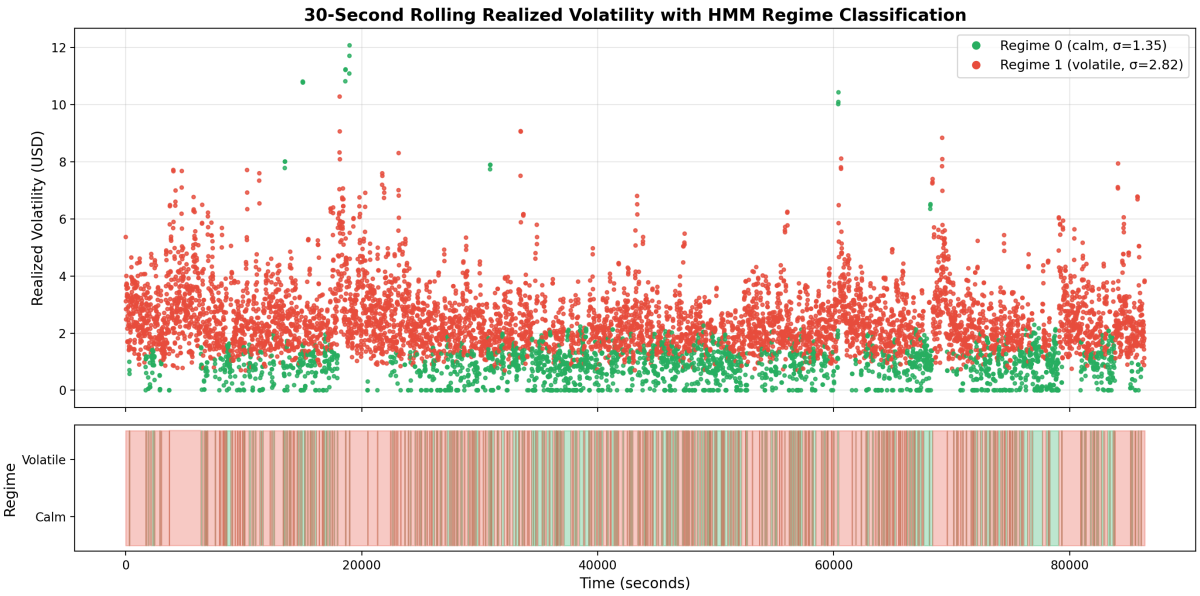


Figure 4.2: Realised volatility (30-second windows) with HMM regime classification.

4.2.2 Two-State HMM Estimation

A two-state Gaussian HMM is fitted to the log of the active realised volatility series (i.e. after filtering) using the Baum-Welch algorithm (Rabiner, 1989). The model assumes that the log realised volatility in each regime is normally distributed, and the latent state follows a first-order Markov chain. The log transform is used because the raw realised volatility is right-skewed, and the Gaussian emission assumption is more appropriate on the log scale.

The estimated parameters are summarised in Table 4.1. The two regimes are well separated: the calm regime ($Y_t = 0$) has a mean log realised volatility of -0.081, and the volatile regime ($Y_t = 1$) has a mean of 0.849. The simulation volatilities σ_i are obtained as the mean raw-scale realised volatility within each inferred regime: $\sigma_0 = 1.351$ USD$/\sqrt{\text{s}}$ (calm) and $\sigma_1 = 2.819$ USD$/\sqrt{\text{s}}$ (volatile). The ratio $\sigma_1/\sigma_0 \approx 2.1$ indicates a meaningful difference in volatility between the two regimes, consistent with the volatility clustering observed in cryptocurrency markets (Ardia et al., 2019; Giudici and Abu Hashish, 2020).

Table 4.1: HMM parameter estimates from 30-second realised volatility.

Parameter	Regime 0 (Calm)	Regime 1 (Volatile)
Mean log(RV)	-0.081	0.849
σ (1s returns, USD)	1.351	2.819
P(stay in regime)	0.827	0.922
Mean duration (seconds)	58	129
Stationary probability	0.309	0.691
Fraction of sample	0.304	0.696

The transition probabilities reveal a moderately asymmetric persistence structure. The volatile regime is more persistent, with a 92.2% probability of remaining in the same state at each 10-second step, corresponding to a mean duration of approximately 129 seconds (about 2.2 minutes). The calm regime has an 82.7% stay probability and a mean duration of about 58 seconds. The stationary distribution places roughly 31% of time in the calm regime and 69% in the volatile regime, producing a reasonable economic split where the market is calm about one-third of the time. The empirical sample fractions (30.4% calm, 69.6% volatile) closely match the stationary distribution, suggesting that the empirical regime frequencies are broadly consistent with the fitted transition structure.

Figure 4.3 shows diagnostic plots: (a) log-RV histograms by regime with fitted Gaussians, showing well-separated clusters; (b) posterior probability of the volatile regime, spending most of the time near 0 or 1 with occasional intermediate uncertainty; (c) conditional return distributions showing wider dispersion and heavier tails in the volatile regime; (d) transition matrix heatmap with high diagonal entries reflecting regime persistence.

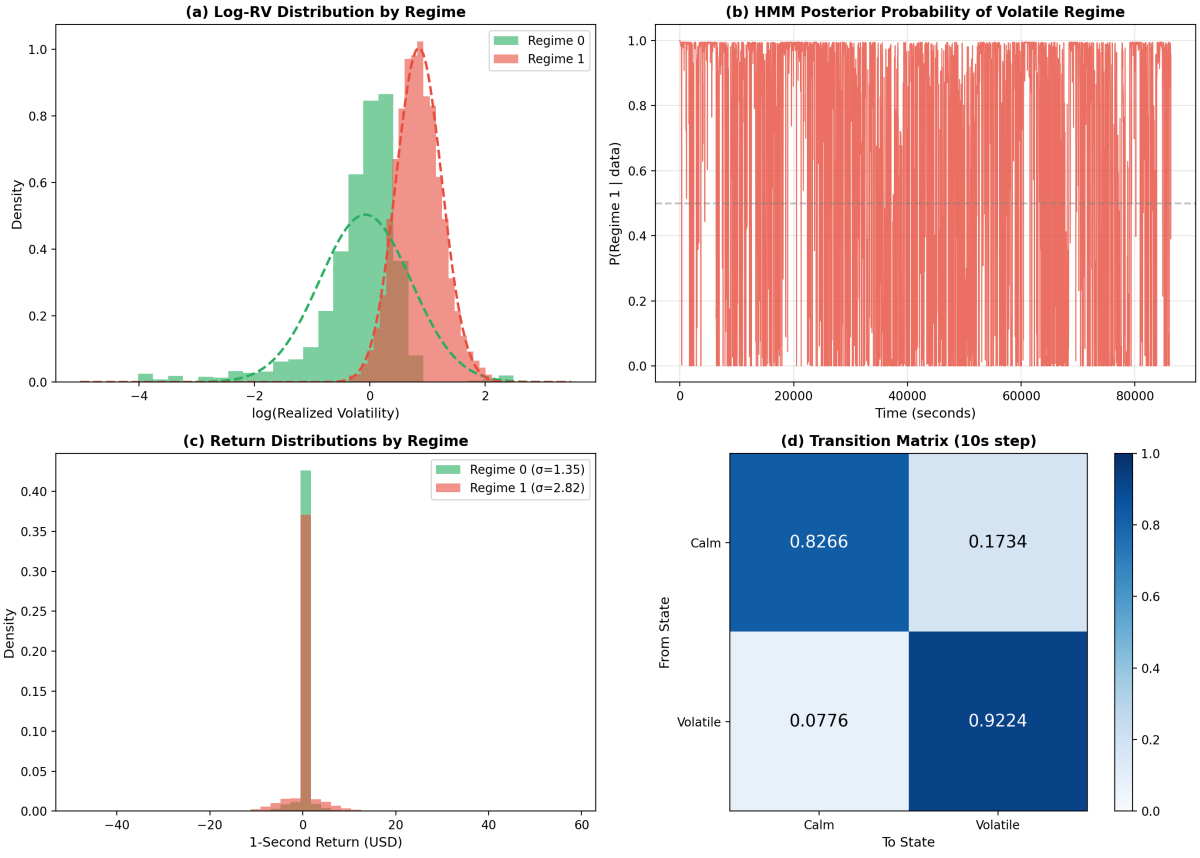


Figure 4.3: HMM diagnostic plots: (a) log-RV distributions by regime with fitted Gaussians, (b) posterior probability of the volatile regime, (c) 1-second return distributions by regime, (d) transition matrix heatmap.

4.3 Intensity Estimation

4.3.1 Estimation Methodology

The order arrival intensity model specifies that the fill rate decays exponentially: $\lambda(\delta) = A\exp(-k\delta)$. This is a reduced-form calibration: we use observed market trades relative to the mid-price as a proxy for the fill intensity that a single market maker would experience at each distance, rather than estimating directly from one market maker's posted quotes. To estimate (A, k) for each regime and side, we compute the distance δ between each trade price and the contemporaneous mid-price, assign trades to regimes via the Viterbi path (Rabiner, 1989) (a hard classification that ignores posterior regime uncertainty, but is a standard and pragmatic simplification), bin by distance, and estimate regime-conditional arrival rates as counts divided by time-in-regime. We then fit the exponential model using weighted least squares on the log scale, with greater weight on bins containing more observations, since sparse bins produce noisier rate estimates.

Figure 4.4 shows the empirical arrival rates and fitted curves. The calm regime exhibits steep decay ($k \approx 1.7$-2.3), with fills concentrated near mid-price, while the volatile regime decays more gradually ($k \approx 0.94$). The volatile regime shows approximate bid-ask symmetry, while the calm regime exhibits noticeable asymmetry in baseline rates.

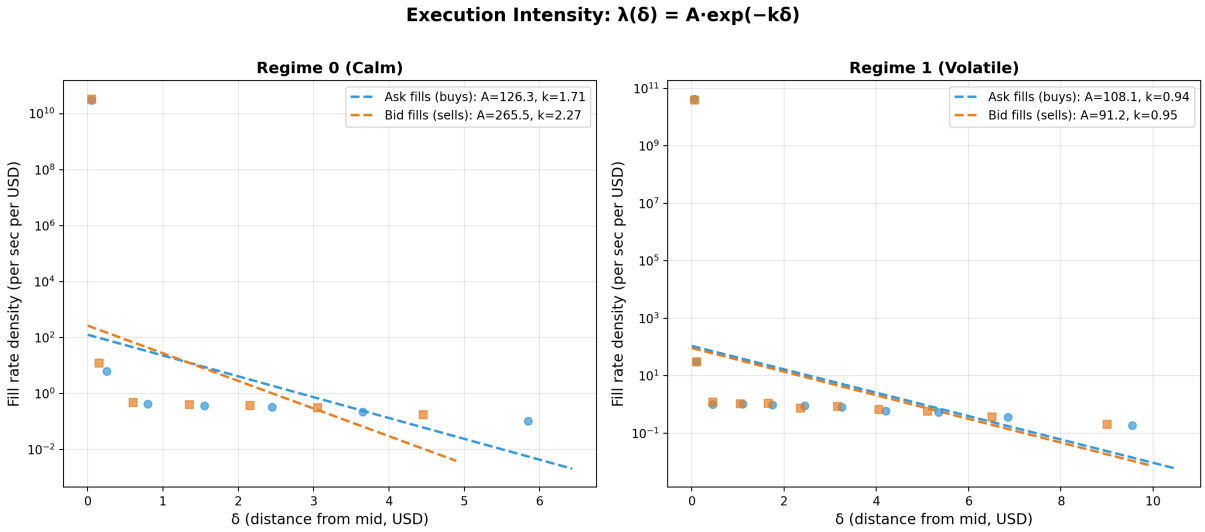


Figure 4.4: Empirical order arrival rates and fitted exponential intensity model. Left: calm regime. Right: volatile regime. Scatter points show binned empirical rates; dashed lines show the fitted $\lambda(\delta) = A\exp(-k\delta)$ curves.

4.3.2 Estimation Results

Table 4.2 reports the intensity estimates. With the filtered HMM approach, the calm regime contains 30.4% of the 30-second windows and approximately 17.1% of all trades, providing sufficient data for direct estimation in both regimes.

Table 4.2: Intensity parameter estimates by regime and side.

Regime	Side	A	k	$1/k$ (USD)
0 (Calm)	Ask (buy)	126.29	1.714	0.583
0 (Calm)	Bid (sell)	265.48	2.273	0.440
1 (Volatile)	Ask (buy)	108.14	0.937	1.067
1 (Volatile)	Bid (sell)	91.25	0.946	1.057

The calm regime has higher baseline rates (A) but much steeper decay ($k_0 \approx 1.7$-2.3 vs $k_1 \approx 0.94$), reflecting tighter liquidity provision. The characteristic spread $1/k$ is \$0.44-\$0.58 in the calm regime versus \$1.07 in the volatile regime. The volatile regime shows approximate bid-ask symmetry, consistent with balanced order flow during high-volatility periods.

4.4 Simulation Parameters

Table 4.3 summarises the full set of parameters used in the Monte Carlo simulations.

4.4.1 Transition Matrix Adjustment

The empirical HMM operates on a 10-second step and produces a stationary distribution $(\mu_0, \mu_1) \approx (0.31, 0.69)$ with mean regime durations of 58 s and 129 s. These durations are far longer than the 10-second simulation horizon, so using the empirical transition matrix directly would produce very few regime switches per trial. We therefore use a deliberately rescaled transition matrix

$$P = \begin{pmatrix} 0.9995 & 0.0005 \\ 0.0010 & 0.9990 \end{pmatrix}, \tag{4.1}$$

with stationary distribution $(2/3, 1/3)$ and mean durations of 2 s (calm) and 1 s (volatile). This creates an illustrative stress-test environment in which regime switches occur frequently enough for the filter to be meaningfully exercised within the short horizon. The volatility levels and intensity parameters remain empirically calibrated; only the transition persistence is rescaled.

Table 4.3: Simulation parameter configuration.

Parameter	Value	Description
Time horizon		
T	10.0 s	Total simulation length
Δt	0.001 s	Time step
Volatility		
σ_0	1.351 USD/$\sqrt{\text{s}}$	Calm regime volatility
σ_1	2.819 USD/$\sqrt{\text{s}}$	Volatile regime volatility
Transition matrix		
p_{00}	0.9995	Calm $\to$ calm
p_{01}	0.0005	Calm $\to$ volatile
p_{10}	0.0010	Volatile $\to$ calm
p_{11}	0.9990	Volatile $\to$ volatile
Ask intensities		
$A_0^+,\ A_1^+$	126.3, 108.1	Baseline arrival rates
$k_0^+,\ k_1^+$	1.714, 0.937	Decay parameters
Bid intensities		
$A_0^-,\ A_1^-$	265.5, 91.2	Baseline arrival rates
$k_0^-,\ k_1^-$	2.273, 0.946	Decay parameters
Market maker parameters		
γ	0.01	Risk aversion
$q_{\max}$	50	Inventory limit
q_0	0	Initial inventory
$\delta_{\min}$	\$0.01	Minimum quote distance
$\delta_{\max}$	\$5.00	Maximum quote distance
Prior belief		
π_0	(0.667, 0.333)	Initial regime belief

4.4.2 Prior Belief

The initial belief is set to $\pi_0 = (0.667, 0.333)$, which matches the stationary distribution of the simulation transition matrix. This means that the market maker begins with a belief that is on average correct but does not know the true starting regime. The naive strategy uses this prior throughout without updating, while the partial information strategy updates it at every time step via the filter from Section 3.2. The oracle strategy ignores the prior entirely, as it observes the true regime directly.

4.4.3 Risk Aversion and Inventory Limits

With $\gamma = 0.01$ and $\sigma_1 = 2.819$, the inventory risk term $\gamma\sigma_1^2\tau \approx 0.795$ at $\tau = 10$, producing meaningful inventory skew without dominating the base spread. The inventory limit $q_{\max} = 50$ prevents extreme positions, and the quote bounds $\delta_{\min} = \$0.01$, $\delta_{\max} = \$5.00$ ensure active quoting throughout.

Chapter 5

Results

This chapter presents the results of the Monte Carlo experiments described in Chapter 4. Section 5.1 compares the aggregate profitability of the three strategies across 200 simulations. Section 5.2 examines how the partial information and oracle strategies adapt their quoted spreads to the prevailing regime. Section 5.3 evaluates the accuracy and convergence speed of the Bayesian regime filter. Section 5.4 decomposes the PnL by regime to understand where the informed strategies gain their advantage. Section 5.5 analyses inventory dynamics, Section 5.6 reports formal statistical tests, Section 5.7 studies the robustness of the results across four different trading days (including a robustness test under the raw empirical transition matrix), and Section 5.8 investigates sensitivity to the risk aversion parameter γ. Section 5.9 synthesises the findings.

5.1 Aggregate PnL Comparison

Table 5.1 summarises the terminal PnL across 200 paired Monte Carlo simulations. All three strategies are profitable under the calibrated-and-rescaled simulation design of Chapters 3-4.

Table 5.1: Monte Carlo PnL summary (200 simulations, 10-second horizon).

Strategy	Mean PnL	Std PnL	Sharpe†	Win vs Naive	Improvement
Naive	\$579.06	\$26.63	21.75	—	—
Partial Information	\$616.04	\$31.63	19.47	83.5%	+6.4%
Oracle	\$621.82	\$34.39	18.08	85.5%	+7.4%

† Cross-simulation mean terminal PnL divided by cross-simulation standard deviation.

The partial information strategy earns a mean terminal PnL of \$616.04, compared to \$579.06 for the naive and \$621.82 for the oracle. It outperforms the naive in 83.5% of

simulations with a 6.4% mean improvement, closing approximately 86% of the gap to the oracle ($36.97/$42.76).

Figure 5.1 visualises the distributions: Panel (a) shows histograms with the informed strategies shifted rightward relative to the naive; Panel (b) plots paired differences centred at approximately $37, with only 16.5% of runs below zero; Panel (c) presents box plots with minimal overlap between naive and informed strategies.

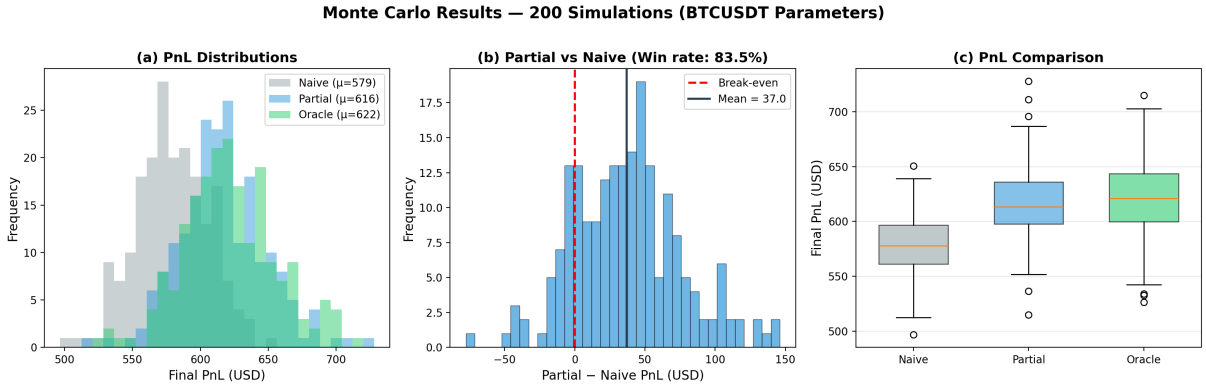


Figure 5.1: Monte Carlo PnL distributions (200 simulations): (a) PnL histograms, (b) partial vs naive paired difference, (c) box plot comparison.

Figure 5.2 shows 20 sample PnL paths. In the first few hundred milliseconds, the paths evolve together while the filter accumulates evidence. Subsequently, the informed (blue) paths pull ahead of the naive (grey), with separation widening steadily, illustrating that the advantage compounds as the filter adapts quotes to the prevailing regime.

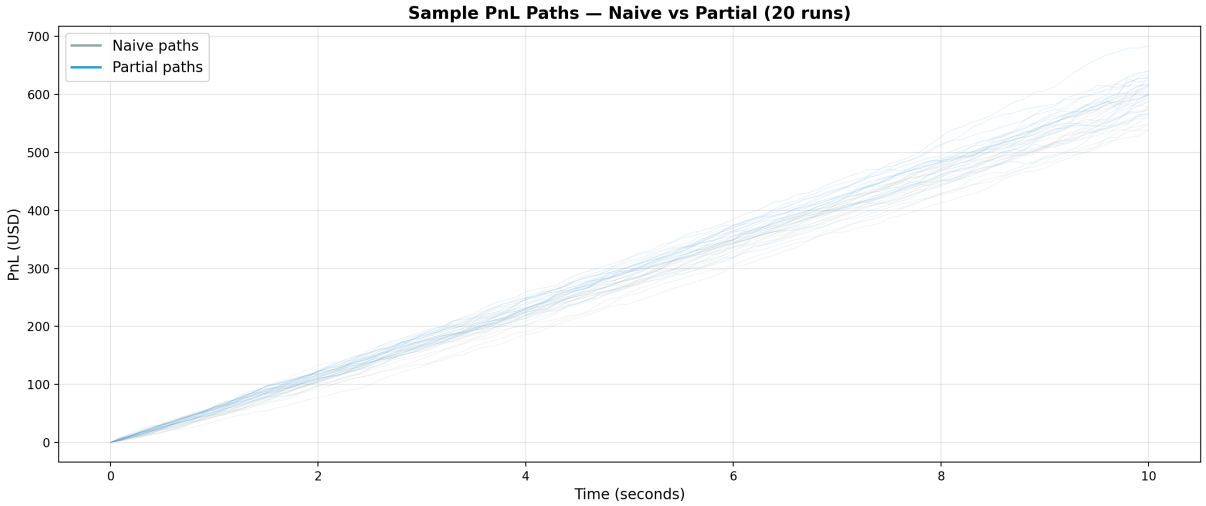


Figure 5.2: Sample PnL paths for naive (grey) and partial information (blue) strategies across 20 simulations.

5.2 Spread Adaptation

The key mechanism through which regime information improves performance is spread adaptation: the informed strategies widen their spreads in the volatile regime (to compensate for greater price and execution uncertainty) and tighten them in the calm regime (to capture more fills when the risk of price movement is lower).

Table 5.2 reports the mean quoted spread by regime for each strategy.

Table 5.2: Mean quoted spread (USD) by regime and strategy.

Strategy	Calm Spread	Volatile Spread	Ratio (Vol/Calm)
Naive	\$1.433	\$1.423	0.99×
Partial Information	\$1.134	\$2.465	2.17×
Oracle	\$1.118	\$2.503	2.24×

The naive strategy quotes nearly identical spreads in both regimes (ratio 0.99×), as it uses unconditional averages throughout. The partial information and oracle strategies both achieve a spread ratio of approximately 2.2×, widening substantially during volatile periods and tightening during calm periods. The partial information strategy nearly matches the oracle (2.17× vs 2.24×), suggesting that the filter provides sufficiently accurate regime estimates.

Figure 5.3 illustrates an example simulation. Panel (a) shows the true regime; Panel (b) shows the naive spread flat at \$1.43, while informed spreads respond sharply to regime switches, jumping to \$2.5 in volatile periods and contracting to \$1.1 in calm periods (oracle reacts immediately; partial information shows a brief filter lag); Panel (c) shows cumulative PnL with informed strategies pulling ahead after volatile episodes.

Figure 5.4 shows spread distributions by regime. Informed strategies tighten to \$1.13 in calm periods (vs naive \$1.43) and widen to \$2.47-\$2.50 in volatile periods (vs naive \$1.42), while the naive quotes remain nearly identical across both regimes.

5.3 Filter Performance

The partial information strategy relies on the Bayesian filter from Section 3.2 to estimate the current regime in real time. If the filter is slow or inaccurate, the strategy cannot adapt its quotes in time and the potential gains from regime information are lost.

Figure 5.5 evaluates the filter on an example run. Panel (a) shows the posterior rapidly converging to high confidence within a few steps of each regime transition, with brief ramps as the filter accumulates observations. Panel (b) shows rolling accuracy consistently

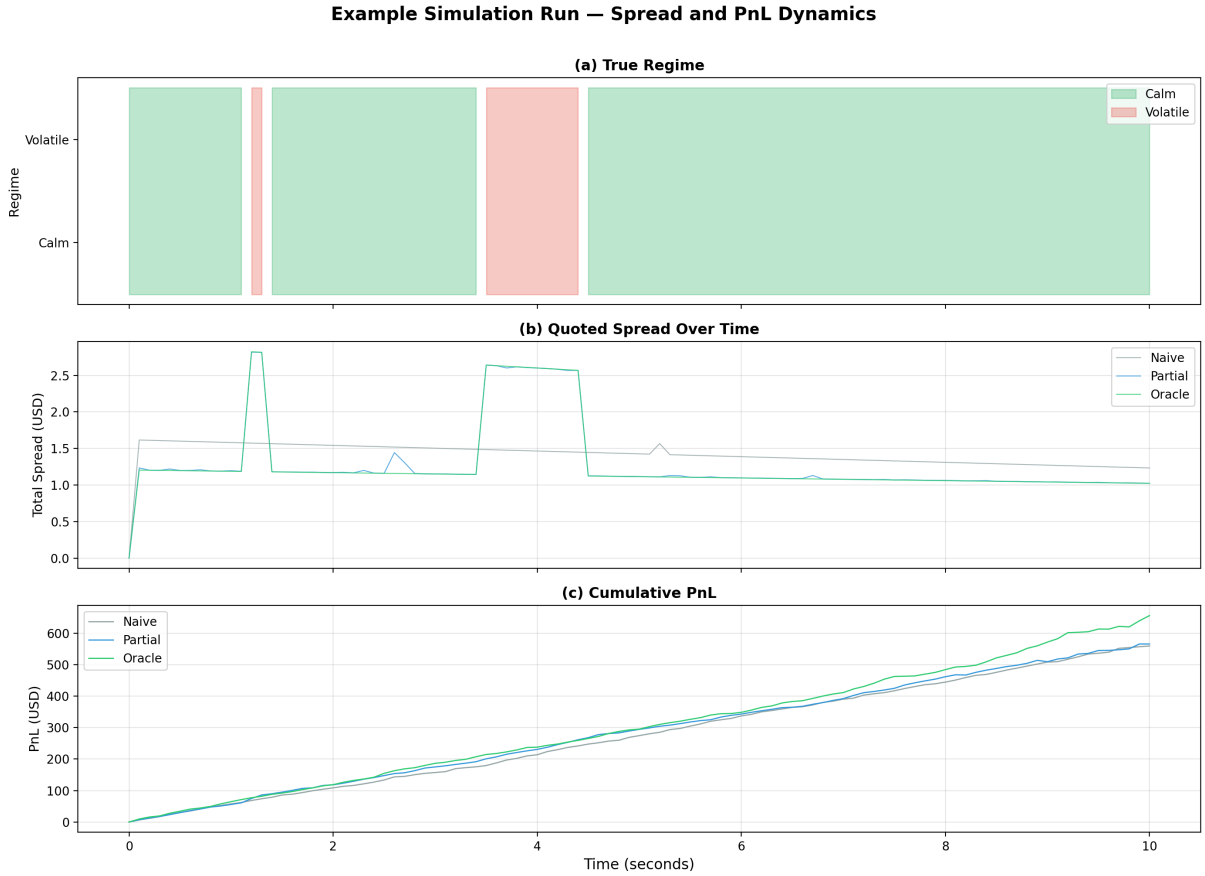


Figure 5.3: Example simulation: (a) true regime, (b) quoted spread over time, (c) cumulative PnL.

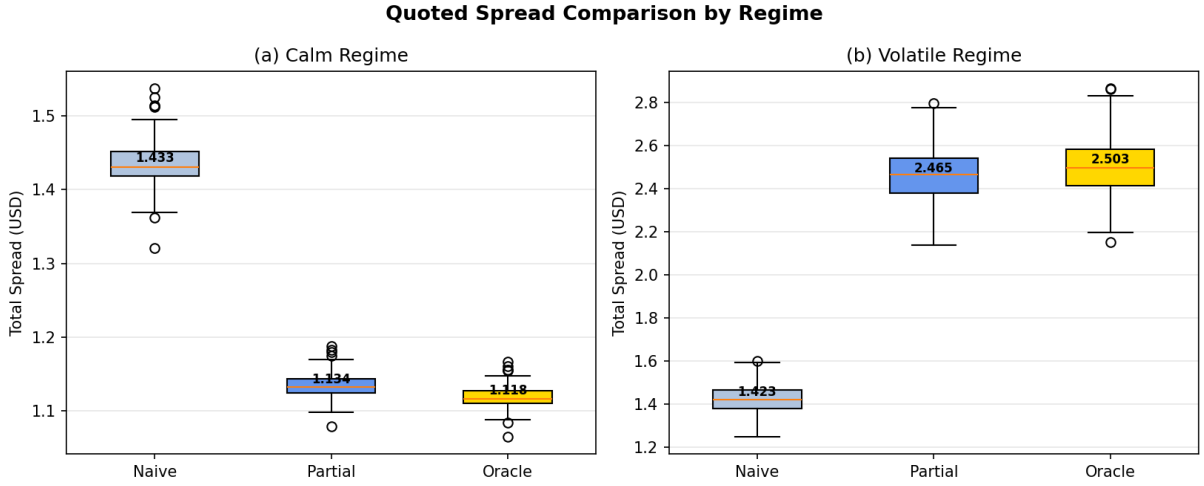


Figure 5.4: Quoted spread distributions by regime: (a) calm regime, (b) volatile regime.

above 97%, with overall accuracy of 99.5%. Figure 5.6 below provides the systematic cross-simulation assessment.

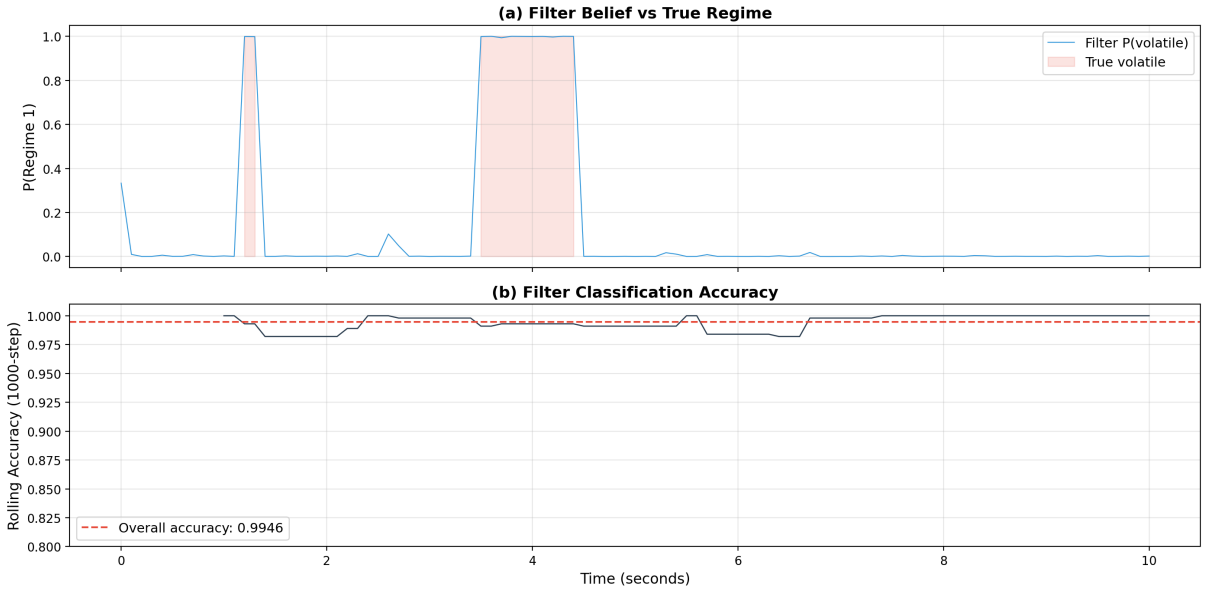


Figure 5.5: Filter performance on an example run: (a) posterior belief vs true regime, (b) rolling classification accuracy.

Figure 5.6 provides the systematic assessment across all 200 simulations. Median convergence to 90% confidence occurs in 15 steps (15 ms at $\Delta t = 0.001$ s), fast relative to 1-2 second regime durations, so the filter spends most of each episode correctly converged. Classification accuracy reaches 99.3% (calm) and 98.5% (volatile), with narrow cross-simulation error bars.

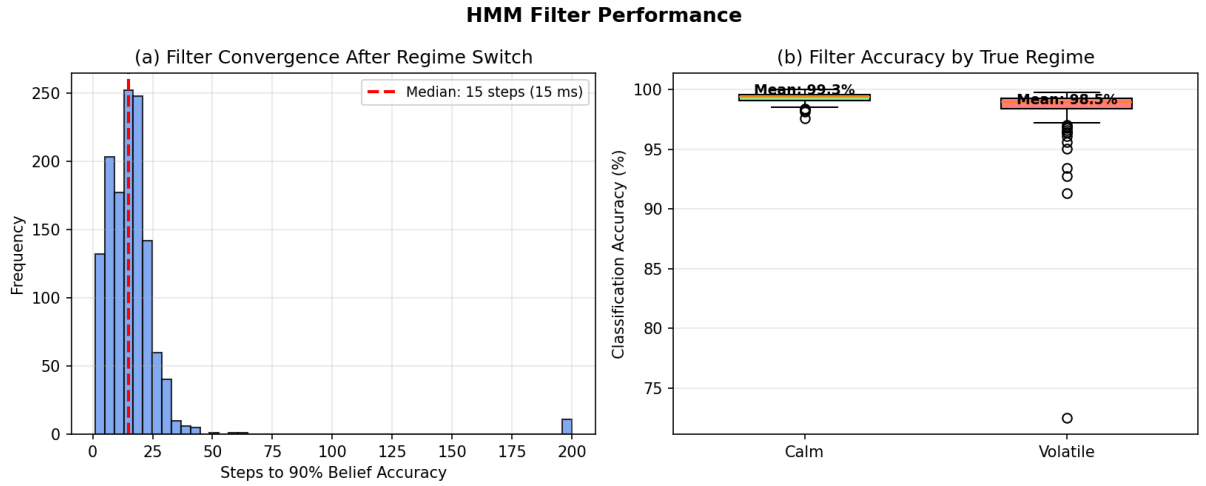


Figure 5.6: Filter convergence across 200 simulations: (a) steps to 90% belief accuracy after regime switch, (b) classification accuracy by regime.

This explains why the partial information strategy performs so close to the oracle: the filter provides highly accurate regime information for most of the simulation.

5.4 PnL Decomposition by Regime

To understand where the informed strategies gain their advantage, we decompose the PnL accrual rate by regime. Table 5.3 reports the mean PnL accrual rate (per 1,000 simulation steps) separately for calm and volatile periods.

Table 5.3: PnL accrual rate by regime (USD per 1,000 steps).

Strategy	Calm Rate	Volatile Rate	Calm % of PnL	Volatile % of PnL
Naive	55.89	62.45	47.2%	52.8%
Partial	59.19	67.62	46.7%	53.3%
Oracle	59.92	69.70	46.2%	53.8%

Improvement is larger in the volatile regime (8.3% vs 5.9% calm), consistent with wider protective spreads being more valuable when price risk is elevated. Figure 5.7 confirms systematic improvement, with error bars overlapping between partial and oracle but not between naive and partial.

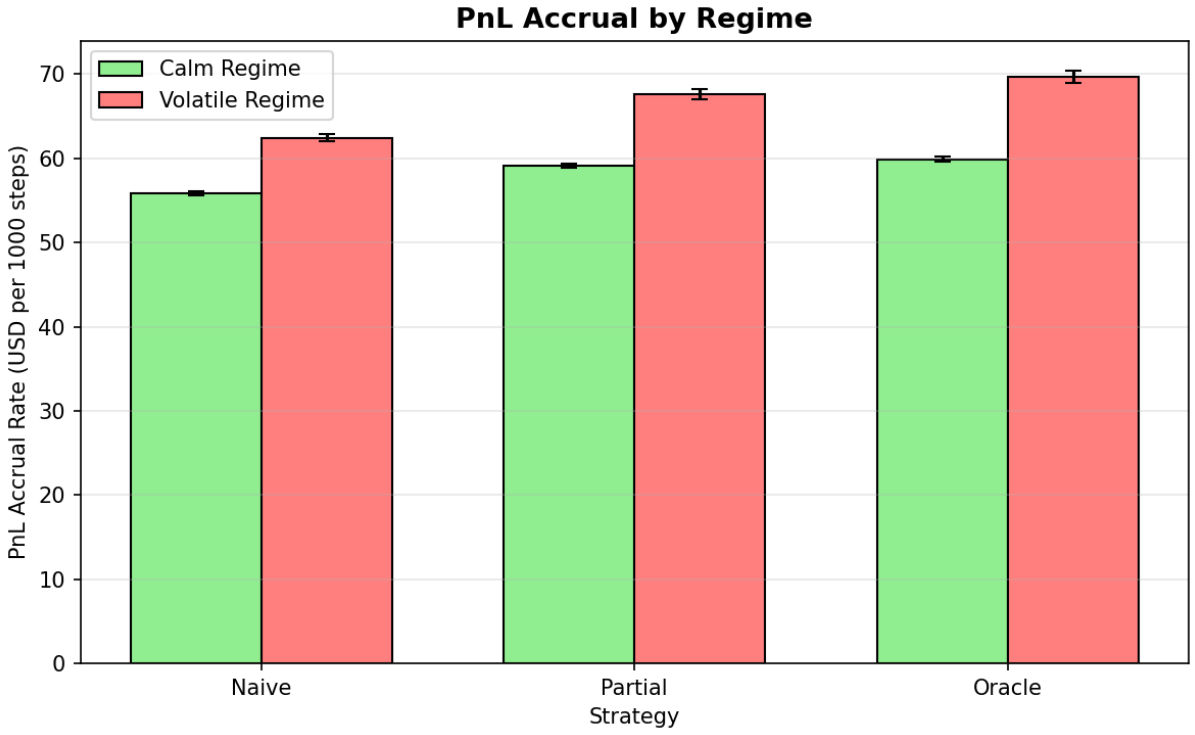


Figure 5.7: PnL accrual rate by regime and strategy (200 simulations, with standard error bars).

5.5 Inventory Analysis

Table 5.4 reports inventory statistics across the 200 simulations.

Table 5.4: Inventory statistics by strategy (200 simulations).

Strategy	Mean $\|q\|$	Std $\|q\|$	Mean Max $\|q\|$	% at Limit
Naive	1.80	0.26	13.8	0.00%
Partial	2.82	0.73	20.9	0.00%
Oracle	2.86	0.74	21.9	0.00%

The informed strategies carry higher average inventory ($|q| \approx 2.8$ vs 1.8 for naive), a natural consequence of tighter calm-regime spreads attracting more fills. Peak inventory remains well within the 50-unit limit (mean max ≈ 21), and no strategy hits the limit. Figure 5.8 shows box plots of the mean and maximum absolute inventory for each strategy. The naive boxes are compact and sit low (mean $|q|$ around 1.8, max $|q|$ around 14), reflecting its wider constant spreads that attract fewer fills. The partial and oracle boxes are nearly identical to each other but sit higher (mean $|q|$ around 2.8, max $|q|$ around 21), with slightly wider interquartile ranges reflecting the greater variability introduced by regime-dependent quoting.

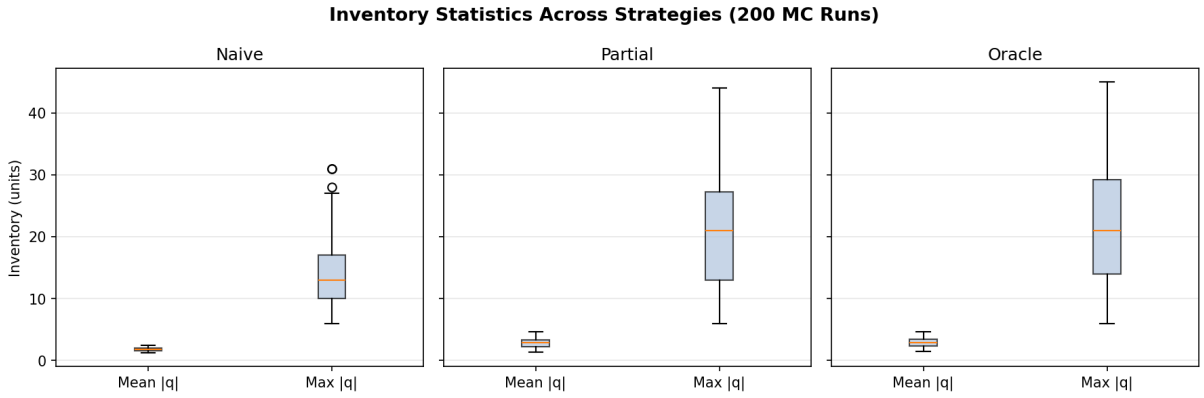


Figure 5.8: Inventory statistics across 200 simulations: mean $|q|$ and max $|q|$ by strategy.

Table 5.5 reports fill rates. The partial information strategy achieves 993 fills per simulation versus 935 for the naive, with the extra fills concentrated in calm periods (782 vs 614).

Table 5.5: Fill rate statistics by strategy (200 simulations).

Strategy	Total Fills	Calm Fills	Volatile Fills	Ask Fills	Bid Fills
Naive	935	614	321	463	472
Partial	993	782	211	489	505
Oracle	998	790	208	491	507

The informed strategies have fewer volatile fills (211 vs 321) but more calm fills (782 vs 614), reducing exposure to heightened price and inventory risk in volatile periods while capturing additional lower-risk calm fills. Although the informed strategies execute fewer trades in volatile periods, these trades occur at wider spreads, so volatile-regime PnL

remains higher (Table 5.3).

5.6 Statistical Significance

Table 5.6 reports formal hypothesis tests for the pairwise PnL differences.

Table 5.6: Statistical significance of pairwise PnL differences (200 paired simulations).

Comparison	Mean Diff	t-stat	p-value	Win Rate
Partial vs Naive	\$36.97	13.57	< 0.0001	83.5%
Oracle vs Naive	\$42.76	14.79	< 0.0001	85.5%
Oracle vs Partial	\$5.78	2.06	0.041	56.0%

The partial-vs-naive improvement is highly significant ($t = 13.57$, $p < 0.0001$), as is oracle-vs-naive ($t = 14.79$, $p < 0.0001$). The paired design controls for path-level variation and reduces path-level noise in the pairwise comparisons. The oracle-vs-partial gap is statistically significant at the 5% level but economically small ($t = 2.06$, $p = 0.041$): the residual gap of \$5.78 per simulation is modest relative to the \$36.97 gain over the naive, indicating that the filter recovers most of the useful regime information.

5.7 Multi-Day Robustness

To assess robustness, the full pipeline is repeated on three additional consecutive days (2024-03-27 to 2024-03-29). Table 5.7 reports the HMM calibrations: the two-state structure is consistently identified, with volatility ratios ranging from 1.9× to 3.0× and calm fractions from 30% to 73%.

Table 5.7: HMM calibration parameters across four trading days.

Date	σ_{calm}	σ_{volatile}	Vol Ratio	Calm Fraction
2024-03-27	4.251	12.614	3.0×	0.730
2024-03-28	3.440	6.546	1.9×	0.343
2024-03-29	2.364	5.326	2.3×	0.477
2024-03-30	1.351	2.819	2.1×	0.304

Table 5.8 reports Monte Carlo results: win rates range from 64.0% to 96.5%, PnL improvements from 6.6% to 18.2%, all statistically significant ($p < 0.01$). The partial information strategy recovers 85-98% of the oracle gap.

The largest improvements occur on higher-volatility days (18.2% on 2024-03-29) and the smallest on the lowest-volatility day (6.6% on 2024-03-30), consistent with regime awareness being most valuable when regimes differ most. Figure 5.9 visualises the cross-day

Table 5.8: Monte Carlo results across four trading days (200 simulations each).

Date	Mean PnL (Naive)	Mean PnL (Partial)	Mean PnL (Oracle)	Win Rate	Improvement	Oracle Gap Closed	p-value
2024-03-27	\$195.04	\$215.34	\$216.26	64.0%	10.4%	96%	0.0008
2024-03-28	\$390.00	\$449.93	\$455.89	86.5%	15.4%	91%	<0.0001
2024-03-29	\$479.00	\$566.33	\$568.35	96.5%	18.2%	98%	<0.0001
2024-03-30	\$579.06	\$616.04	\$621.82	83.5%	6.4%	86%	<0.0001
Cross-day mean				82.9%	12.7%	92%	

results, with the partial information strategy exceeding the naive and nearly matching the oracle on every day.

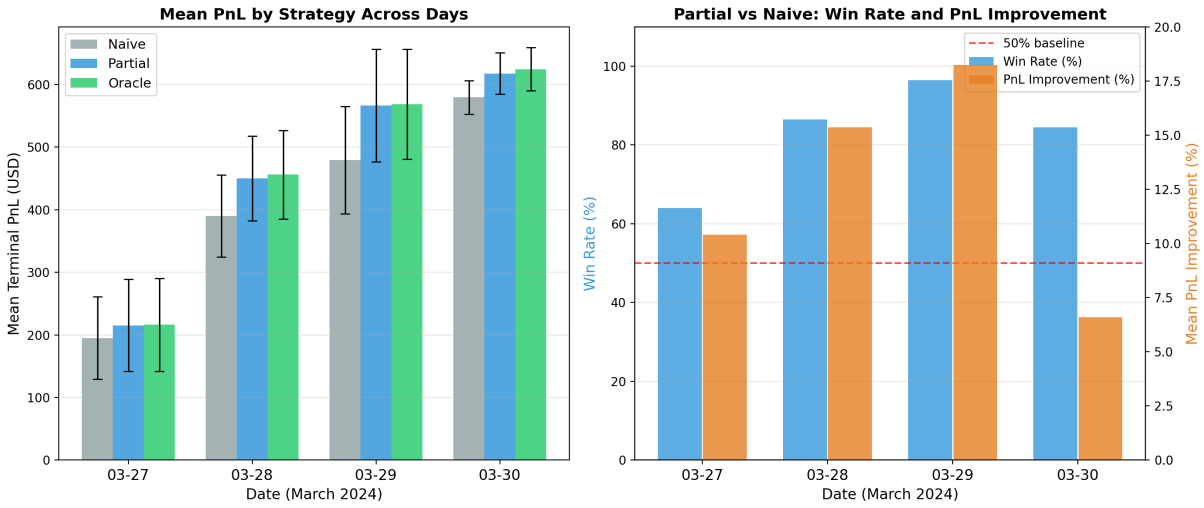


Figure 5.9: Cross-day robustness: (a) mean PnL by strategy, (b) win rate and PnL improvement of partial vs naive.

Figure 5.10 shows calibrated volatility parameters: absolute levels vary across days, but the two-regime structure is consistently identified with volatility ratios ranging from 1.9× to 3.0× (mean 2.3×).

These results indicate that the advantage is robust across day-specific calibrations with meaningfully different volatility conditions.

5.7.1 Robustness to the Empirical Transition Matrix

The main experiments use a rescaled transition matrix P_{sim} (Section 4.4.1) that compresses regime durations to 1-2 seconds, creating frequent switches within the 10-second horizon. A natural question is whether the partial information advantage persists under the *raw empirical* transition matrix, which has much longer regime durations (approximately 50 s for calm, 111 s for volatile).

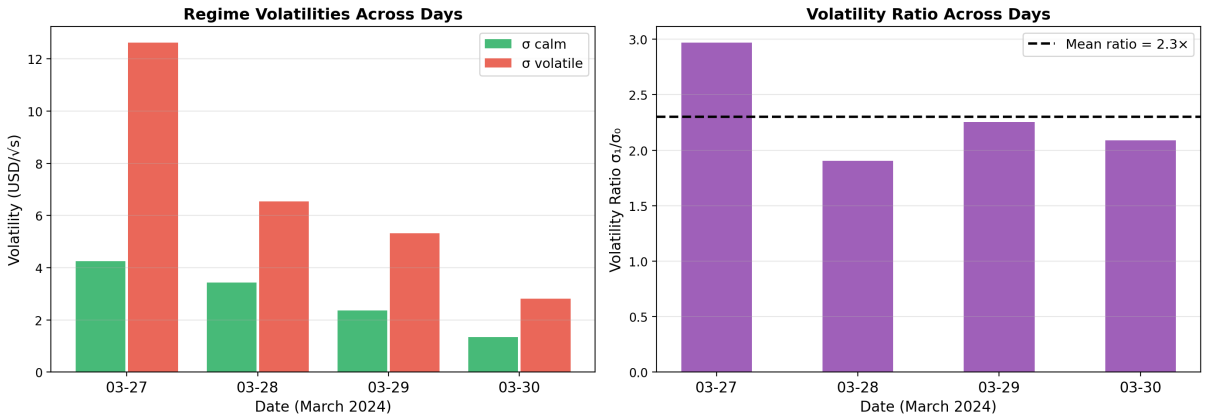


Figure 5.10: Cross-day volatility parameters: (a) regime volatilities, (b) volatility ratio.

Table 5.9 reports the results of 200 paired simulations using the empirical matrix P_{emp} derived from the 2024-03-30 HMM calibration via matrix logarithm conversion to the $\Delta t = 0.001$ s step. All other parameters (volatilities, intensities, γ) are identical to the main experiment.

Table 5.9: Monte Carlo results under the empirical (non-rescaled) transition matrix (200 simulations).

Strategy	Mean PnL	Std PnL	Win vs Naive	Improvement	t-stat
Naive	\$608.14	\$35.86	—	—	—
Partial Information	\$663.42	\$48.38	91.5%	+9.1%	17.61
Oracle	\$664.20	\$50.37	91.5%	+9.2%	17.03

Under the empirical matrix, 177 of 200 runs contain zero regime switches within the 10-second horizon, and the mean number of switches is 0.1. The initial regime is drawn from the empirical stationary distribution (31% calm, 69% volatile), so most simulations run entirely within a single regime. Despite this, the partial information strategy achieves a *higher* win rate (91.5% vs 83.5%) and improvement (9.1% vs 6.4%) than the main rescaled experiment. This occurs because, under the empirical matrix, the filter quickly converges to near-certainty about the (typically constant) regime, and the entire simulation benefits from correctly adapted quotes. The near-zero gap between partial and oracle (\$0.78) confirms that with stable regimes the filter provides essentially perfect information. These results show that the advantage of regime-aware quoting is not an artefact of the artificially frequent switching in the rescaled design; it persists, and is in fact amplified, under empirical transition persistence.

5.8 Sensitivity to Risk Aversion Parameter

The results presented above use a fixed risk aversion parameter $\gamma = 0.01$. Since γ governs the trade-off between inventory risk and execution aggressiveness (Section 3.3), it is important to verify that the advantage of the partial information strategy is not an artefact of a particular γ choice. To assess this, the full 200-run paired Monte Carlo experiment is repeated for five values of γ spanning 1.5 orders of magnitude: $\gamma \in \{0.001, 0.005, 0.01, 0.02, 0.05\}$.

Table 5.10 reports the results. The partial information strategy outperforms the naive at every γ tested, with all improvements statistically significant ($t > 2.8$, $p < 0.01$).

Table 5.10: Sensitivity of strategy performance to the risk aversion parameter γ (200 paired simulations, 2024-03-30 calibration).

γ	Mean PnL (Naive)	Mean PnL (Partial)	Mean PnL (Oracle)	Win Rate	Improvement	t-stat
0.001	\$618.46	\$644.68	\$640.63	57.5%	+4.2%	2.80
0.005	\$600.30	\$634.94	\$635.75	76.5%	+5.8%	8.88
0.010	\$579.06	\$616.04	\$621.82	83.5%	+6.4%	13.57
0.020	\$542.08	\$586.86	\$586.00	92.5%	+8.3%	19.72
0.050	\$447.33	\$505.14	\$503.70	87.5%	+12.9%	18.82

Two patterns emerge. First, the improvement increases with γ: from 4.2% at $\gamma = 0.001$ to 12.9% at $\gamma = 0.05$. This is consistent with the theoretical structure of the quoting rule: as γ increases, errors in regime-dependent risk assessment become more costly, since the reservation price adjustment scales with $\gamma\sigma^2_{\text{eff}}$ and the spread component depends on γ through both the volatility and intensity terms. A more risk-averse market maker places greater weight on correctly estimating the prevailing regime, so the penalty for using unconditional averages is larger. Second, absolute PnL decreases with γ (wider spreads, fewer fills), but the relative advantage is robust, with win rates above 57% and all improvements statistically significant ($t > 2.8$, $p < 0.01$).

5.9 Synthesis

The results across the preceding sections trace a single causal chain from filter quality to PnL improvement. The Bayesian filter achieves classification accuracy above 99% with a median convergence time of 15 steps (Section 5.3), providing the partial information strategy with near-real-time regime estimates. These estimates drive spread adaptation (Section 5.2): the informed strategy widens its quoted spread by 2.17× in the volatile regime and tightens it in the calm regime, closely tracking the oracle's 2.24× ratio. The

adapted spreads reshape the fill composition (Table 5.5): calm-regime fills increase from 614 to 782 per simulation while volatile fills decrease from 321 to 211, shifting the portfolio of executed trades toward lower-risk periods. This shift in fill composition, combined with wider per-trade margins during volatile periods, produces the aggregate PnL gain of 6.4% on the primary day (Section 5.1). The decomposition by regime (Section 5.4) confirms that the volatile regime contributes the larger proportional improvement (8.3% vs 5.9%), consistent with spread adaptation being most valuable when price and inventory risk are elevated. The robustness analyses, across four trading days (Section 5.7), under the empirical transition matrix (Section 5.7.1), and over a range of γ values (Section 5.8), all reinforce this mechanism rather than revealing alternative explanations.

Chapter 6

Discussion and Conclusion

This chapter interprets the simulation results from Chapter 5, discusses the limitations of the approach, identifies directions for future work, and concludes the dissertation.

6.1 Summary of Findings

This dissertation extended the Avellaneda-Stoikov framework by introducing a latent Markov chain that jointly governs mid-price volatility σ and order-arrival intensity parameters, controlling both inventory and execution risk channels simultaneously. The Monte Carlo experiments calibrated to BTC-USDT perpetual futures data demonstrate statistically significant improvements: the partial information strategy outperforms the naive in 83.5% of simulations on the primary day (6.4% mean improvement), with a cross-day mean win rate of 82.9% and 12.7% improvement.

Within the simulation environment, the partial information strategy recovers 86% of the oracle gap on the primary day and 93% on average across four days, with filter classification accuracy above 99% and median convergence of 15 steps (15 ms at $\Delta t = 0.001$ s). The market maker does not need perfect regime knowledge; a rapidly converging belief suffices.

6.2 Interpretation of Results

6.2.1 The Mechanism of Improvement

Spread adaptation drives the improvement: the informed strategy widens spreads 2.2× in volatile regimes (nearly matching the oracle's 2.24×), reducing volatile fills (211 vs 321 naive) while increasing calm fills (782 vs 614 naive). This shifts fill composition toward low-risk periods. Table 5.3 confirms larger volatile-regime improvement (8.3% vs 5.9% calm), reflecting the higher cost of ignoring volatility when price risk is elevated. Although informed strategies execute fewer volatile trades, these occur at wider, more profitable spreads.

6.2.2 The Value of Joint Regime Dependence

The joint dependence of σ and (A, k) on a single hidden chain is the central modelling contribution of this dissertation. In contrast to Campi and Zabaljauregui (2020), where the hidden chain governs only the intensity parameters and the volatility σ is constant, our model allows the Bayesian filter to use two information channels: the price channel (large price increments signal the volatile regime via the likelihood ratio in Equation (3.12)) and the event channel (fill patterns signal the regime through the intensity model). Both channels are informative and typically reinforce each other, which helps explain the filter's rapid convergence.

From the quoting perspective, the joint model also means that the belief π_t simultaneously adjusts both the reservation price (through the effective volatility σ^2_{eff}) and the quoted spread (through the belief-weighted intensity parameter k_{eff}). This two-channel adjustment is more powerful than either alone: when the filter detects a shift to the volatile regime, the reservation price moves to reflect higher inventory risk while the spread widens to reflect the changed fill probability structure. The result is a quoting rule that adapts more rapidly and more effectively to changing market conditions. A natural extension would be to isolate the marginal contribution of each channel by running a volatility-only variant (holding intensity parameters constant across regimes), which would directly quantify how much of the improvement derives from the intensity channel versus the volatility channel.

6.2.3 Cross-Day Variation

The cross-day results are consistent with a simple intuition: regime awareness is most valuable when the regimes differ most. Days with higher volatility ratios (up to 3.0×) show larger improvements (up to 18.2%), while the lowest-volatility day shows the smallest improvement (6.6%). Despite volatility ratios ranging from 1.9-3.0× and calm fractions from 30-73%, the partial information strategy consistently outperforms the naive, suggesting that a two-state regime description is robust across the sampled BTC-USDT trading days.

6.3 Limitations

Several limitations of the analysis should be acknowledged. These are discussed below, organised into modelling assumptions, calibration choices, and simulation design.

6.3.1 Modelling Assumptions

The model inherits the Avellaneda-Stoikov assumptions: arithmetic Brownian mid-price (no drift), exponential intensity model, and CARA utility. The no-drift assumption is reasonable over the 10-second horizon but becomes questionable at longer horizons where momentum or mean-reversion may be relevant. The exponential intensity $\lambda(\delta) = A\exp(-k\delta)$ is tractable but potentially misspecified, with possible non-zero intercept at $\delta = 0$ or heavier tails.

The two-state regime assumption is a modelling simplification. Cryptocurrency markets may exhibit more than two volatility states, and a three-state or continuous-volatility model could potentially capture a richer set of market conditions. However, the two-state model has the advantage of parsimony: it introduces the fewest additional parameters while still capturing the fundamental distinction between calm and volatile periods.

6.3.2 Calibration Choices

The HMM calibration on 30-second windows involves a trade-off: longer windows improve volatility precision while shorter windows detect rapid transitions. Alternative windows (10s or 60s) could yield different classifications. The two-state structure is robust across days, though window-length sensitivity is not directly tested.

The order-arrival intensity parameters are estimated from trade data using a simple exponential fit to the empirical fill rate as a function of distance from the mid-price, conditioning on the HMM-inferred regime. This is a reduced-form approach that does not account for the endogeneity of the market maker's quoting strategy: in a live market, the observed fill rate depends on the strategies of all market participants, not just the exponential decay assumption. The estimated parameters should therefore be interpreted as calibration inputs for the simulation, rather than as structural estimates of order flow dynamics.

6.3.3 Simulation Design

The Monte Carlo simulation is a controlled laboratory setting, not a live trading environment. Several simplifications are worth noting. First, while the regime volatilities and intensity parameters are empirically calibrated, the transition matrix used in the simulations is deliberately rescaled (Section 4.4.1) to produce regime switches frequent enough for the filter to be meaningfully exercised within the 10-second horizon. The raw empirical transition matrix has regime durations of approximately one minute, during which a 10-second window would typically contain at most one transition; the rescaled matrix compresses regime durations to 1-2 seconds, creating a stress-test environment. This means the simulation results demonstrate that the filtering mechanism works under favourable switching conditions, but do not directly characterise performance under empirical transition persistence. Second, the simulation assumes that the market maker is the only strategic agent: the mid-price follows an exogenous process, and other market participants do not adjust their behaviour in response to the market maker's quotes. In a real market, other participants would adapt, potentially eroding the advantage of the regime-aware strategy. Third, the simulation does not include transaction costs, latency, or partial fills, all of which would reduce the realised PnL in practice. Fourth, the 10-second horizon is very short; a longer horizon would introduce additional complications such as inventory management over multiple regime cycles, overnight risk, and the possibility of model parameters drifting over time.

Finally, the Monte Carlo design generates both the regime path and the price process according to the calibrated model, which means that the simulation is testing the value of regime awareness *within the model*, not in a fully out-of-sample sense. If the true data-generating process differs substantially from the assumed HMM structure, the benefits of filtering could be smaller. The multi-day robustness analysis partially addresses this concern by showing that the advantage persists under different calibrations, but a definitive test would require live or paper trading on future data.

6.4 Directions for Future Work

Several natural extensions suggest themselves, grouped into modelling, methodological, and empirical directions.

Online parameter learning. The current implementation holds HMM and intensity parameters fixed within each simulation day, recalibrating only across days. An online extension using recursive expectation-maximisation (Cappé, 2011) or particle filtering to jointly estimate regimes and parameters within a single trading session would improve adaptivity, particularly for cryptocurrency markets where microstructure properties can shift on the scale of hours. This would also remove the need for a pre-calibration window, making the strategy more practical for live deployment.

Richer state spaces. The two-state model could be extended to three or more regimes, or to a continuous latent volatility process (e.g. a stochastic volatility model with regime-dependent drift), capturing intermediate market conditions. A $K > 2$ extension would require assessing whether the additional regime granularity justifies the increased parameter count and potential overfitting risk.

Incorporating drift and alpha signals. The current model assumes a driftless mid-price. Incorporating short-term drift or alpha signals (e.g. from order flow imbalance or trade-arrival asymmetry) into the regime-switching framework would allow the market maker to tilt quotes directionally, potentially improving performance in markets with predictable short-horizon price movements.

Realistic market simulation. The Poisson arrival model is a tractable simplification. A more realistic simulation could replay historical limit order book data, exposing the strategy to empirical queue dynamics, partial fills, and latency. This would also allow testing under adverse selection, where a fraction of incoming orders are informed and predict future price movements.

Transaction costs and latency. The simulation does not account for exchange fees, network latency, or the possibility of quotes being stale when they are executed. Including these frictions would reduce the absolute PnL of all strategies, and it would be valuable to quantify whether the *relative* advantage of regime-aware quoting survives under realistic cost structures.

Live trading validation. The most important next step is testing in a live or paper trading environment, providing a genuine out-of-sample test against real-world complications such as latency, partial fills, and strategic adaptation by other participants. The Binance testnet provides a natural venue for this validation.

6.5 Conclusion

This dissertation has developed and tested a regime-aware market-making strategy extending the Avellaneda-Stoikov framework with joint regime dependence of volatility and intensity parameters. The partial information strategy achieves a mean win rate of 82.9% and 12.7% PnL improvement across four trading days, recovering 86% of the oracle's advantage on the primary day and 93% on average across four days.

The core insight is that a rapidly converging Bayesian filter, fed by both price and event observations, provides sufficiently accurate regime estimates for near-optimal quoting. The spread adaptation mechanism, widening during volatile periods and tightening during calm ones, drives the improvement by reducing exposure to heightened price and inventory risk in volatile periods while capturing more lower-risk calm fills. The advantage is robust across γ values and increases with risk aversion, consistent with the structure of the quoting rule.

The central theoretical contribution is that coupling volatility and intensity under a single latent chain creates a strictly more informative filtering problem than governing either parameter alone: the two observation channels (price increments and order executions) reinforce each other, enabling near-oracle quoting from a lightweight predict-update recursion. The formal results (Propositions 3.3 and 3.5) ground this claim, and the simulation evidence across multiple days, transition regimes, and risk-aversion levels confirms its practical relevance for cryptocurrency market making.

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